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Multicriteria Analysis of Quality of Life in European Union Countries

ΛΑΜΠΡΟΥ ΘΕΟΔΩΡΟΣ του ΙΩΑΝΝΗ

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με επιβλέποντα

ΠΑΠΑΘΑΝΑΣΙΟΥ ΙΑΣΩΝΑ

Υποβλήθηκε ως απαιτούμενο για την απόκτηση του διπλώματος μεταπτυχιακών σπουδών στην Αναλυτική των Επιχειρήσεων και Επιστήμη των Δεδομένων

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To Ioannis, Niki, and Konstantinos Lamprou, for their courage, love and support through everything up until today.

Preface

This thesis was done due to the motivation I had from both a personal and academic interest in the factors that play vital role in Quality of Life. Personal interest derived from the fact that I want to pursue a professional career abroad, therefore a study showing results about the well-being of several European Countries would help me make a more data-driven decision. My academic interest on the other hand emerged due to how complex Quality of Life has become nowadays and its many aspects. The tool used in this research, PROMETHEE II method, is extremely significant in exploring Quality of Life and its indicators. I am deeply grateful to my supervisor, Jason Papathanasiou, who guided me to successfully complete my thesis. Special thanks to all those that participated voluntarily to the questionnaire made about the significancy of the indicators we chose for Quality of Life. Their answers provided a view that was interesting to observe and compare. This thesis reflects great understanding on how data can support decision-making at both societal and individual level.

Abstract

The main purpose of this thesis is to evaluate Quality of Life in a variety of European Union countries selected for examining. The method used is the PROMETHEE II multicriteria decision analysis. The data for our subject were collected from Eurostat, covering a 10-year period from 2014-2023 for ten EU members. Quality of Life has a complex definition and is composed of many factors. We chose 10 key criteria, reflecting several aspects of well-being such as economic performance, housing quality, social vulnerability and others.

A Python implementation was performed to rank Quality of Life among the EU countries. For this script, 2 scenarios were introduced. The first one involves an aggregated analysis using the average values for each criterion across the 10-year range for every country. The second consists of a yearly evaluation where each country will be compared even with itself but for different year, for example Denmark of 2020 with Denmark of 2019 etc. The PROMETHEE II algorithm was fully constructed within Python, providing each calculation step and even enabling us to visualize it in excel especially for the average scenario due to less data. This can be done for the yearly method as well but due to much more data it will be more complex and difficult to read. Additional cross-validation was performed using Visual PROMETHEE software. This helped in comparing the Python results for the aggregated scenario with the ones that derive from the Visual PROMETHEE program. Furthermore, beyond the standard equal-weighted scenario, a questionnaire was distributed to 100 university students and graduates to capture their opinion on how important each criterion is. Using the results of this survey alternative weights were formed for further investigation and correlation with the researcher's scenario. This research offers valuable insights into cross-country differences in Quality of Life across the European Union Countries over the last 10 years.

Keywords: Quality of Life, Multicriteria Analysis, PROMETHEE, Eurostat, European Union

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1. Introduction

Quality of Life is a concept that has many aspects. Some of the key factors are economic prosperity, social well-being, health, education, housing conditions and more. For someone to measure Quality of Life it would require more than just looking at income or Gross Domestic Product. Multiple domains should be considered simultaneously, capturing their significance in the overall well-being.

In the European Union, the concept of Quality of Life is crucial among members. European Union over the years increasingly focuses on improving citizens living standards across several criteria. Therefore, it becomes vital to assess how countries perform in all those indicators. Because of the multi-dimension Quality of Life has, there is need for tools that can work on data from sources like Eurostat and comprehend their effect on the overall well-being. This research will apply the PROMETHEE II method, a well-established multi criteria decision analysis (MCDA) technique, to evaluate Quality of Life in selected European Union countries and rank them altogether. The overall aim is to offer a data-driven approach on well-being differences between countries in the European Union and how much the indicators chosen affect each country's performance. The usefulness of MCDA methods thus will also be demonstrated. This analysis hopes to inform about the living standards and challenges faced across different European Union countries over the past 10 years.

2. Literature Review

In this section, the concepts that are fundamental to our research will be introduced. We begin by describing definitions of Quality of Life as established in the literature and the factors that usually describe it as a concept. The importance of Multi-Criteria Decision Analysis Methods in multidimensional topics such as well-being, will be discussed. Later, a view of previous studies, that have used MCDA methods in evaluating Quality of Life, will be provided. Emphasis is given also to the PROMETHEE and Visual PROMETHEE tools. Since the criteria have a vital role in the PROMETHEE methodology, a set of indicators and their significance will be introduced. Furthermore, a brief presentation of existing Quality of Life rankings in the European Union is provided. Finally, the objective of this research is defined along with the research questions that will help us later in the analysis of our subject.

2.1 Quality of Life Definitions

Quality of Life according to Teoli and Bhardwaj (2025) aims to capture the total well-being either of a population or a person. Positive and negative elements are both taken in consideration. Quality of Life includes a variety of indicators with some of them being according to Teoli and Bhardwaj (2025) "personal health (physical, mental), relationships, education, work environment, social status, wealth, physical and economic security and physical surroundings".

The experts of Eurostat (2017) explain QoL as "a broader concept than only economic production and living standards." The full range of factors that most people value in their lives according to Eurostat (2017) is:

1. "Material living conditions
2. Productive or other main activity
3. Health
4. Education
5. Leisure and social interactions
6. Economic security and personal safety
7. Governance and basic rights
8. Natural and living environment

9. Overall experience of life”

(Eurostat, 2017)

According to Teoli and Bhardwaj (2025) the World Health Organization claims that QoL is an evaluation of a person’s perception of their reality compared to their goals. The University of Toronto similarly addresses QoL as the ability to enjoy meaningful life opportunities. QoL should not be confused with similar terms like standard of living, that refer to financial status or health related indicators about QoL that specifically focus on health’s impact on the overall well-being. Since a universal definition hasn’t been made, researchers have started categorizing QoL into separate dimensions. “Engaged Theory” is an approach that organizes QoL into four different domains:

1. Economics
2. Culture
3. Politics
4. Ecology

According to OECD (2024), Better Life Index, “There is more to life than the cold numbers of GDP and economic statistics – This Index allows you to compare well-being across countries, based on 11 topics the OECD has identified as essential, in the areas of material living conditions and quality of life”.
According to OECD (2024) “The indicators used to measure Quality of Life are:

1. Housing
2. Income
3. Jobs
4. Community
5. Education
6. Environment
7. Civic Engagement
8. Health
9. Life Satisfaction
10. Safety
11. Work-Life Balance”

(OECD, 2024)

2.2 MCDA Methods

Multi Criteria Decision Analysis according to Zlaugotne, Zihare, Balode, and Kalnbaļķīte (2020) is an approach that involves several steps and methods designed to make decision-making clear, consistent and transparent. Over time, a variety of MCDA methods and software tools have been developed to handle various decision-making problems. To effectively use MCDA, it's crucial first to define the problem clearly, to identify the alternatives and choose relevant criteria. These criteria should include factors specific to the situation.

When multiple alternatives to evaluate exist, MCDA helps determine the most suitable option based on several factors. Different MCDA methods can lead to different rankings even when using the same data, because each method has its own logic. Methods are chosen based on the type of outcome needed. For instance, methods like AHP, MULTIMOORA, MAUT, Weighted Product Method are useful when comparing values. Alternatively, methods such as AHP, TOPSIS, VIKOR, COPRAS, and STEP can be used to identify the best choice among several options. PROMETHEE and ELECTRE methods are based on pairwise comparisons to assess suitability according to chosen criteria.

Having similar results among MCDA methods though isn't impossible. Studies have found that there is possibility in gaining the same results while using different methods. Using the same input data for all five MCDA methods while different results were produced, similar trend to the best alternative existed. Although, because results are similar but not the same, it isn't wise to compare results that derive from different methods.

2.3 Quality of Life Previous Research

Studies in the past years have attempted to evaluate Quality of Life using Multi-Criteria Decision Analysis methods, each trying a different perspective in their methodology.

In the study "The Classification of Quality of Life Using Multi-criteria Analysis", Massam (1999) criticizes the Simple Additive Weighting (SAW) method, emphasizing the challenges that occur when trying to determine appropriate weights for each indicator. To solve this limitation, the authors applied the Aspiration-level Interaction Method (AIM) that enables ranking without requiring defined weights. This research has a similar goal to the one that the present study has since it tries to evaluate Quality of Life between 2 individuals. The

thing that differs is that we also compare Quality of Life between individuals, but ones that come from different regions of Europe since we test the well-being across European Union Countries.

A different focus is observed in "A Multi-Criteria Weighting Approach for Quality of Life Evaluation", where according to Rădulescu et al. (2019), accurate weighting of indicators is emphasized. These indicators are mostly related to health, psychological, social relationships and the environment. This view of Quality of Life differs from the present research, which observes Quality of Life from another field. This field includes economic conditions, material deprivation, physical safety and more. This contrast shows how flexibles MCDA methods are and how complex the concept of Quality of Life is.

In the study "Multi-Criteria Decision Making to Measure Quality of Life: An Integrated Approach for Implementation in the Urban Area of Thessaloniki, Greece", Feneri et al. (2015) examine the concept of Quality of Life but focus in the scope of a city instead of a country, which is what happens in the present research. Analytic Hierarchy Process, being a famous MCDA method, is used to observe urban living conditions. Their research is based on both subjective and objective criteria. For the subjective criteria a questionnaire was performed with participants answering questions about indicators that were made from the researchers. The objective criteria derived from Eurostat and were traditional Quality of Life indicators. The indicators chosen from the authors resemble the ones used in the present research.

Urban-level analysis is also seen in the case of Tabriz according to Ahadnejad Reveshty and Hazeri (2020), in "Quality of Life Indicators Associated with Urban Prosperity Case Study: Tabriz City", where Quality of Life within a city is investigated. The approach is different though, since they focus on how better QoL is inside different regions of the city. Their work applies a combination of Analytic Hierarchy Process and PROMETHEE. The indicators derived from the City Prosperity Index framework. They reflect economic development, environmental conditions, transportation and public services. AHP is used to determine the weights of each indicator and then PROMETHEE is used to generate the rankings. Some of the criteria used are similar to the ones used in the present research and also there is not much difference in the methodology since in the Tabriz City Case Study, different regions of a city are compared using the PROMETHEE method while in the

present research different countries.
(Ahadnejad Reveshty and Hazeri, 2020)

Another contribution is "A Hybrid Multiple-Attribute Decision-Making Model with Modified PROMETHEE for Identifying Optimal Performance-Improvement Strategies for Sustainable Development of a Better Life", where according to Hu and Tzeng (2019), a decision-making model designed to identify strategies for improving Quality of Life is presented. Several MCDA methods are used with some of them being DEMATEL, DANP and a modified version of PROMETHEE. The OECD Better Life Index is used, including both Material living condition criteria and other Quality of Life factors such as housing, income and employment. This research shows how PROMETHEE can guide decision makers extremely well. It is relevant to the present study since it applies PROMETHEE in a well-being context.
(Hu and Tzeng, 2019)
More technical innovation is seen in "Measuring quality of life in Europe: A new fuzzy multicriteria approach", where according to Reig-Mullor et al. (2024), Quality of Life is observed through the use of fuzzy sets. They argue that traditional income-based approaches are too limited and the concept of well-being is so complex that it needs multidimensional view. Their methodology is based on the use of picture fuzzy sets which is an extension of fuzzy logic. This model is applied to assess Quality of Life across European Union countries. This work is relevant to the present research since it measures Quality of Life but using a different methodology. Many of the criteria are similar to the ones used in the present study with some of them being the same, but the authors using fuzzy sets used a greater number of indicators comparing to ours.
(Reig-Mullor et al., 2024)

In other researches, such as "A Geographic Visualization Approach to Multi-Criteria Evaluation of Urban Quality of Life" according to Rinner (2007), Analytic Hierarchy Process is used to calculate Quality of Life rankings across different neighborhoods in Toronto.

In "Development of The Multi-Criteria Model for the Quality of Life Assessment in Local Governments" according to Mijailović and Marković (2022), Quality of Life is examined across areas by using both objective and subjective criteria. Individual preferences, life goals, expectations and satisfaction are all indicators that shape Quality of Life. The research takes into account different age, sec, education and social status. The MCDA

method used for evaluation is PROMETHEE combined with descriptive statistics. This research resembles the present study since PROMETHEE is used to evaluate Quality of Life but differs since in the present study variations about age, gender or education aren't taken into account and the overall Quality of Life of every individual in a country is found.

Furthermore, in "Quality of Life in the European Union: A Multidimensional Analysis" Rogge and Van Nijverseel (2019) analyze Quality of Life across European Union countries. While the concept is exactly the same compared to the present study, a different approach in measuring Quality of Life since the geometric Benefit of the Doubt (BoD) method is used to create a well-being index. The results revealed higher well-being scores in Nordic and Western European countries compared to Southern and Eastern which is also findings that occurred in the present study as-well.

Finally, in "The Economic Crisis and its Effects on the Quality of Life in the European Union", Somarriba Arechavala et al. (2015) examined the impact of the 2007-8 economic crisis on the Quality of Life in the European Union countries by comparing indicators from 2007 and 2011. Their goal is in exploring the extent of the negative impact of the economic crisis in measuring a citizen's well-being. A cross-country comparison was done and the goal was to see which countries declined in well-being and which remained stable. This research is directly relevant to the present study as it shows the huge impact economic conditions have in Quality of Life even though PROMETHEE is not used, and the methodology includes using a synthetic indicator.

Together, these studies confirm the vital role MCDA methods like PROMETHEE and others have in evaluating Quality of Life, since it is a complex concept that needs multidimensional analysis. Our research builds on this idea by using a country-level PROMETHEE II method across a decade, that has many perspectives since weights are defined from both the researcher and weights that derive from participants of a questionnaire.

2.4 PROMETHEE Methodology

The main structure of this section, the PROMETHEE methodology is followed by the detailed explanation of PROMETHEE II methodology in Papathanasiou and Ploskas (2018). As of April 2017, approximately 1500 studies referencing this method were cataloged by

Professor Bertrand Mareschal, showing the method's significancy.
Input requirements according to Papathanasiou and Ploskas (2018) are similar to the ones used in methods like TOPSIS and VIKOR, although PROMETHEE allows users to have greater flexibility by including additional variables based on a variety of preference functions. PROMETHEE was later enhanced by GAIA (Geometrical Analysis for Interactive Aid), a module used for visualization and designed to represent more complex decision problems, graphically in a two-dimensional format. While GAIA is of great importance for analysts due to its visual clarity, it typically requires a solid mathematical understanding. The PROMETHEE method will be given an in-depth analysis while the GAIA will be briefly introduced.

PROMETHEE actually represents a collection of methods, since it has had many extensions and improvements over time. The primary approaches range from PROMETHEE I through VI. PROMETHEE produces rankings of different alternatives that are given the name of "actions" based on their degree of preference. Usually this involves comparing pairs of actions across each criterion chosen individually, computing preference flows per criterion, and then combining these individual flows into a total, global ranking. PROMETHEE has proven effective in a variety of application domains, such as nuclear waste management, agricultural productivity, risk analysis, website evaluation, renewable energy planning, and more.

PROMETHEE based on Brans and Mareschal (2005) "is designed to deal with multicriteria problems such as the following :

$$\max\{g_1(a), g_2(a), \dots, g_n(a) \mid a \in A\}$$

where A is a finite set of possible alternatives $\{a_1, a_2, \dots, a_m\}$ and $\{ g_1(\cdot), g_2(\cdot), \cdots, g_n(\cdot)\}$ a set of evaluation criteria that are either maximized or minimized." (Brans and Mareschal, 2005) The decision maker needs to construct an evaluation Table as the following:

a	$g_1(\cdot)$	$g_2(\cdot)$	$\cdots$	$g_n(\cdot)$
	w_1	w_2	$\cdots$	w_n
a_1	$g_1(a_1)$	$g_2(a_1)$	$\cdots$	$g_n(a_1)$
a_2	$g_1(a_2)$	$g_2(a_2)$	$\cdots$	$g_n(a_2)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
a_m	$g_1(a_m)$	$g_2(a_m)$	$\cdots$	$g_n(a_m)$

Figure 1: Evaluation Table

According to Papathanasiou and Ploskas (2018) "MCDA methods in general put the decision makers at the heart of the process, allowing them to shape the analysis according to their own preferences. These methods serve only for aid and do not replace the human judgement since the final responsibility lies with the decision maker. In PROMETHEE specifically, preferences between actions are represented through preference degrees that range from 0 (no preference) to 1 (absolute preference). If differences between criteria evaluations are minor, decision makers usually assign small preference values while when these differences are negligible, PROMETHEE can capture this as well. When large differences are found, higher preference scores occur and when they surpass a threshold determined by the decision maker, one alternative is considered absolutely preferable over another." (Papathanasiou and Ploskas, 2018)

Therefore, when a criterion is to be maximized, the preference function is defined as:

$$P_j(a, b) = F_j[d_j(a, b)], \forall a, b \in A$$

And the preference degree as we said before is always in (0,1)

$$0 \leq P_j(a, b) \leq 1$$

When indicator g must be minimized, then -g has to be maximized.

According to Papathanasiou and Ploskas (2018) "The pair $\{g_j(\cdot), P_j(a, b)\}$ is defined as generalized criterion associated to criterion $g_j(\cdot)$". The total number of preference functions that exist are 6. The purpose of them is for the decision maker to have 6 options on how to prefer one action over the other.
Papathanasiou and Ploskas (2018) also mention that:

"In order to model whether the difference in the evaluations among some of the actions in a single criterion is negligible or denotes an absolute preference among a pair of actions, the decision maker has to input at most some variables. These variables are the threshold of indifference q and the threshold of absolute preference p. Q is the threshold below which there is no preference to either of the actions meaning preference degree is 0. (Papathanasiou and Ploskas, 2018) P is the threshold above which there is a total preference to one of the two actions and therefore preference degree is 1. In the case

though of a Gaussian function one more variable exists which is the S. This one is an intermediate value between q and p".

According to Brans and Mareschal (2005):

"in case of a Gaussian criterion the preference function remains increasing for all deviations and has no discontinuities, neither in its shape, nor in its derivatives. A parameter s has to be selected; it defines the inflection point of the preference function. We then recommend to determine first q and p and to fix s in between. If s is close to q the preferences will be reinforced for small deviations, while close to p they will be softened." (Papathanasiou and Ploskas, 2018) While the introduction of the thresholds seems to add subjectivity to the method, it allows more flexibility in the modeling of the specific problem. The only time the thresholds are not needed is when the decision maker chooses the usual criterion. (Brans and Mareschal, 2005)

The **six types of preference functions** according to Papathanasiou and Ploskas (2018):

1. **"Usual Criterion:** This criterion requires no parameters q, p, or s to fix indicating that even the smallest deviation among a pair of actions is taken into account in the final ranking. The preference degree takes either the value of 0 or 1. (Papathanasiou and Ploskas, 2018) In case of 0 there is indifference between the actions showing that they are both considered equally preferred. In case of 1 there is an absolute preference of one action over the other. (Papathanasiou and Ploskas, 2018) This criterion is the simplest but also useful when the decision maker can't decide for the p and q values. Using this criterion made the PROMETHEE method fully comparable with TOPSIS and VIKOR since it does not require in this case the definition of the additional variables we introduced (q, p, or s). (Papathanasiou and Ploskas, 2018)
2. **U-shape Criterion:** This criterion requires only the parameter q. The decision maker is able to create a value for the indifference threshold. (Papathanasiou and Ploskas, 2018) As we said before, below this threshold, actions are indifferent to the decision maker. (Papathanasiou and Ploskas, 2018) However, even the smallest value above the q parameter is crucial since the preference

function instantly takes the value of 1. The function acts like a binary variable taking either 0 or 1. (Papathanasiou and Ploskas, 2018)

3. **V-shape Criterion:** This criterion requires only the parameter p. (Papathanasiou and Ploskas, 2018) The value of the preference function now can range between 0 and 1. In the case of the deviation being above the p threshold it acquires the value of 1. (Papathanasiou and Ploskas, 2018)
4. **Level Criterion:** This criterion requires both p and q parameters. (Papathanasiou and Ploskas, 2018) When the deviation is below q the preference degree is 0 and when its greater than p it is equal to 1. (Papathanasiou and Ploskas, 2018) Any value between the thresholds means that the preference degree is equal to 0.5. Therefore, the preference function can only take the values 0, 0.5, 1. (Papathanasiou and Ploskas, 2018)
5. **V-shape with indifference Criterion:** This criterion is also known as linear. (Papathanasiou and Ploskas, 2018) It is the same with the Level criterion but with the difference that the preference degree does not have a fixed value between the thresholds, being a linear function acquiring any value between 0 and 1. (Papathanasiou and Ploskas, 2018)
6. **Gaussian Criterion:** This criterion needs the decision maker to fix only the parameter s. (Papathanasiou and Ploskas, 2018) If the deviation is less than or equal to 0, then the preference degree is 0. If on the other hand its greater than 0 then the preference degree has the value of 1-e$^{((-d^2)/(2s^2))}$". (Papathanasiou and Ploskas, 2018)

The Preference Functions and how to use them are seen below:

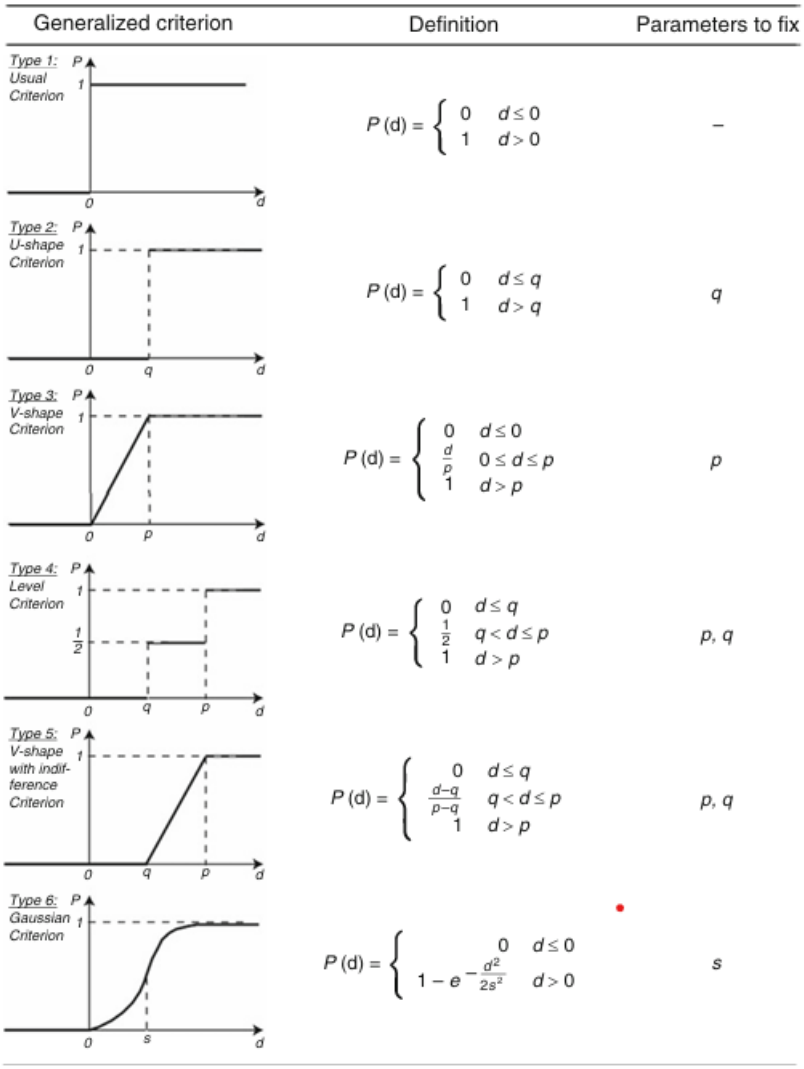


Figure 2: Preference Functions

Papathanasiou and Ploskas also mention that:

"The aggregated preference indices can be calculated as follows:

$$\pi(a, b) = \sum_{j=1}^{n} Pj(a, b)Wj$$

$$\pi(b, a) = \sum_{j=1}^{n} Pj(b, a)Wj$$

Where (a, b) ∈ A, and π(a, b) indicates how much action a is preferred to b over all of the criteria, while π(b, a) is the exact opposite meaning how much action b is preferred to a. (Papathanasiou and Ploskas, 2018) For all (a, b) ∈ A the following properties hold true:

$$\pi(a, a) = 0$$
$$0 \leq \pi(a, b) \leq 1$$
$$0 \leq \pi(b, a) \leq 1$$
$$0 \leq \pi(a, b) + \pi(b, a) \leq 1"$$

According to Papathanasiou and Ploskas (2018):

"When having m actions, each action is competing against (m-1) other, in the set A. The unicriterion positive flow of each action in the set, is a number between 0 and 1 and is an indicator of how much this action is preferred over all the other actions in the set A. The higher this value is, the more preferable this action becomes to the decision maker." (Papathanasiou and Ploskas, 2018) The definition for the positive outranking flow is:

$$\Phi^{+}(a) = \frac{1}{m-1} \sum_{x \in A} \pi(a, x)$$

According to Papathanasiou and Ploskas (2018):

"The negative outranking flow is the exact opposite from the positive outranking flow. It is an indicator of how all the other actions are preferred over this exact action. The definition for the negative outranking flow is:

$$\Phi^{-}(a) = \frac{1}{m-1} \sum_{x \in A} \pi(\chi, \alpha)$$

PROMETHEE I and PROMETHEE II methods have differences according to the authors of PROMETHEE. PROMETHEE I provides the decision maker with a partial ranking of actions while PROMETHEE II gives a complete one. The second method seems to have more popularity in the literature.

PROMETHEE I needs both Φ+ and Φ− rankings to compute the final ranking. When these 2 flows conflict, the actions are considered to be incomparable."

Furthermore:

$$\begin{cases} aP^I b \text{ iff } \begin{cases} \phi^+(a) > \phi^+(b) \text{ and } \phi^-(a) < \phi^-(b), \text{ or} \\ \phi^+(a) = \phi^+(b) \text{ and } \phi^-(a) < \phi^-(b), \text{ or} \\ \phi^+(a) > \phi^+(b) \text{ and } \phi^-(a) = \phi^-(b); \end{cases} \\ aI^I b \text{ iff } \phi^+(a) = \phi^+(b) \text{ and } \phi^-(a) = \phi^-(b); \\ aR^I b \text{ iff } \begin{cases} \phi^+(a) > \phi^+(b) \text{ and } \phi^-(a) > \phi^-(b), \text{ or} \\ \phi^+(a) < \phi^+(b) \text{ and } \phi^-(a) < \phi^-(b); \end{cases} \end{cases}$$

Figure 3: Preference, Indifference and Incomparability

P, I, and R stand for preference, indifference, and incomparability, respectively.

PROMETHEE II is based on the net flow Φ, which is defined as

$$\Phi(a) = \Phi^+(a) - \Phi^-(a)$$

As the net flow value get larger, the action is better.

$$\begin{cases} aP^{II}b \text{ iff } \phi(a) > \phi(b) \\ aI^{II}b \text{ iff } \phi(a) = \phi(b) \end{cases}$$

Figure 4: Net Flows

Incomparabilities do not exist in PROMETHEE II. Even though PROMETHEE II is more frequently used, the first version also has positive aspects and should be considered in using. Net flow can be calculated like

$$\phi(a) = \phi^+(a) - \phi^-(a) = \frac{1}{m-1} \sum_{j=1}^{n} \sum_{x \in A} \left[P_j(a,x) - P_j(x,a) \right] w_j$$

Figure 5: Net Flow Definition

$$\phi(a) = \sum_{j=1}^{n} \phi_j(a) w_j$$

Figure 6: Net Flow 2nd Definition

Where the weights w_j, j = 1, 2, …, n are also considered.
The steps of the PROMETHEE algorithm can be summarized as follows according to Papathanasiou and Ploskas (2018): "

1. Decision matrix construction and setting the values of the preference parameters and the weights.
2. Calculation of the deviations between the evaluations of the alternatives in each criterion.
3. Calculation of the pairwise comparison matrix for each criterion.
4. Calculation of the unicriterion net flows.
5. Calculation of the weighted unicriterion flows.
6. Calculation of the global preference net flows.
7. Ranking of the actions according to PROMETHEE I or PROMETHEE II rules."

2.5 Visual PROMETHEE

According to Papathanasiou and Ploskas (2018):

> "There are not many software packages available for PROMETHEE today. PROMCALC was first introduced from the writers during the 80s. It is a software package that runs under MSDOS. It then evolved to Decision Lab 2000 with a visual interface and a full documentation. This tool is no longer supported though. Some other alternatives that exist are Smart Picker Pro and DECERNS, both of which require a license though. Visual PROMETHEE, developed by Professor Bertrand Mareschal, luckily exists though, and is provided as an academic edition free of charge".

According to Mareschal (2013):

"Visual PROMETHEE is a multicriteria decision aid software. It helps in:

1. Evaluating a number of decisions or items according to a number of conflicting criteria.
2. Identifying the best possible decision.
3. Ranking possible decisions from best to words.

4. Sorting items into predefined classes.
5. Visualizing decision or evaluation problems
6. Achieving decisions when decision-makers have conflicting points of view.
7. Justifying or invalidating decisions."

According to Mareschal (2013):

"Visual PROMETHEE is developed by VPSolutions under the supervision of Professor Bertrand Mareschal from the Solvay Brussels School of Economics and Management of the Universite Libre de Bruxelles (ULB). According to Mareschal (2013), using Visual PROMETHEE one can share the knowledge of a worldwide expert in the field of multicriteria decision aid and of one of the original authors of the PROMETHEE and GAIA methods. (Mareschal, 2013)

Some key terms will be explained bellow according to Mareschal (2013):

1. **Action**: This term is used to designate either a possible decision or an item to evaluate. The program compared different alternatives that are judged on several criteria.
   Synonyms: alternative, decision, item. (Mareschal, 2013)
2. **Category:** This term defines a subset of actions that all share the same symbol and colors in the analysis made. (Mareschal, 2013)
3. **Criterion**: This term defines an attribute that is related to each action and makes it possible to compare all actions and determining the best ones. It can either be quantitive or qualitive. (Mareschal, 2013)
4. **Weight**: This term defines the weight of a criterion which is a positive number that shows the importance of each criterion. The weights of all the criteria in total must sum to 1. In Visual PROMETHEE they are automatically normalizes so that the sum is exactly 1 but the user can enter any positive number as a weight. (Mareschal, 2013)

In Visual PROMETHEE there are many visualization options and tables that help us in our research. (Mareschal, 2013)

1. **The PROMETHEE rankings**: Two rankings are computed. The first is the PROMETHEE I Partial Ranking which is based on the computation of the two

preference flows Phi+ and Phi-. (Mareschal, 2013) The second is the PROMETHEE II Complete Ranking which is based on the net preference flow (net flow). (Mareschal, 2013)

2. **The PROMETHEE diamond**: It is an alternative two-dimensional joint representation of both the versions of PROMETHEE. (Mareschal, 2013) The square corresponds to the Phi+,Phi- plane where each action is represented by a point. The angle of the plane is 45° so that the Phi net flow is given from the vertical dimension. Phi+ scores increase from the left to the top corner and Phi- scores from the left to the bottom corner. (Mareschal, 2013)
3. **The PROMETHEE network**: It is a representation of the PROMETHEE I Partial Ranking that seems similar to the older PROMETHEE software PromCalc or Decision Lab. (Mareschal, 2013) Actions are represented by nodes and arrows and are drawn to show preferences, meaning that incomparabilites are easy to detect. The network representation is like a close-up of the Diamond view where preferences are indicated by arrows. (Mareschal, 2013)
4. **The PROMETHEE table**: It displays the Phi, Phi+ and Phi- scores. The actions are ranked according to the second version of PROMETHEE. (Mareschal, 2013) This helps when having a huge number of actions or in exporting the data to another program such as Excel. (Mareschal, 2013)
5. **The GAIA plane**: It is a standard 2D analysis. GAIA starts from a multidimensional representation of the problem we have with as many dimensions as the number of criteria we use. (Mareschal, 2013) The mathematical method Principal Components Analysis is used to reduce the number of dimensions while not losing information. 3 dimensions are computed. The first is U which contains the maximum possible quantity of information. The second is V which provides the maximum additional informational orthogonal to U. The third one is the W which provides the maximum additional information orthogonal to both U and V. U and V is covered from the standard GAIA analysis. (Mareschal, 2013) When the U, V representation quality is not high enough, he third component W is added from Visual PROMETHEE to improve the analysis. The quality level is displayed at the bottom right of the window of the program. When having a quality level above or close to 70% then the 2D GAIA analysis is reliable. In the representations

we have: Actions represented by points, Criteria represented by axes and the weighting of the criteria and the PROMETHEE II ranking are represented by the decision axis."

(Mareschal, 2013)

2.6 Quality of Life Indicators

In this subsection, key definitions related to the concept of Quality of Life are presented. Definitions, data and information in this QoL Indicators subsection derive mainly from the Eurostat (2017) report "Final report of the expert group on Quality of Life indicators". Eurostat has a wide variety of data and exclusively a section about Quality of Life. These definitions offer an understanding of various dimensions that shape Quality of Life, including economic conditions, environmental quality, physical safety and more. Based on these definitions and the indicators available through Eurostat, we selected a set of relevant criteria to be used in the application of our PROMETHEE methodology. This allows for a structured comparison of Quality of Life across European Union countries, ensuring that our analysis is both data-driven and also reflective of the multifaceted nature Quality of Life has in modern societies.

1) Material Living Conditions: Being available to possess material goods has long been tied to the advancement of societies. In economics, this perspective is evident in growth theory, with its primal concern being in the changes in Gross Domestic Product (GDP). However, since this view is mostly material-focused, it is often reduced to monetary indicators, a fact that has faced criticism. In this day and age, there is work in development towards a more complex understanding of what Quality of Life means. Policies that focused only on economic growth are no longer enough to comprehend the concept of well-being. Material living standards though, remain key part in understanding Quality of Life. According to Eurostat (2017), the Stiglitz Commision acknowledged this by naming material conditions as the first of eight Quality of Life dimensions. Gross Domestic Product remains also as one of the core of the well-being indicators since it captures the overall economic activity within a specific territory.

Capturing deprivation in both income and housing is highly relevant to Quality of Life because of its impact on people's lives. Indicators of Material Deprivation are designed to show a household's living conditions. Eurostat devides Material Living Conditions in 3

subsections, them being Income, Consumption and Material Condition.
(Eurostat, 2017)

A) Income: The Canberra Handbook, 2001 (according to Eurostat, 2017) states that income is considered the most reliable indicator (or the least problematic) of Quality of Life. Consumption and wealth are also crucial for understanding economic well-being since they are part of the Income indicator. While individuals meet their needs and achieve their goals due to their income, wealth supports the sustainability of these choices.
Increases in income are many times linked to better outcomes in a variety of aspects of Quality of Life such as longer life expectancy and higher educational attainment. Economic resources in a country enables it to invest in key services for the Quality of Life such as health, public safety, natural environment and more. So the connection between well-being and Income is positive.
For household income, an indicator used is median income, which focuses on absolute income levels, while another is the at-risk poverty rate which examines inequality and poverty. Because the median is unaffected by extreme values, hence the median income is unaffected by outlier income values. This provides a more reliable reflection of the standard household's financial situation. This means that using the median instead of the mean is more trustworthy since averages can be skewed by outliers.

- **C.1 Median equivalized disposable income**

Definition:
According to Eurostat (2017):

> "The median represents the middle value in a set of numbers that is arranged in an order. It devides the group into two equal halves with half the values being above it and the other half being below it, being unaffected by extreme values.
>
> Household disposable income is computed by summing all monetary income received by every household member. This includes wages, investments and any other income received. From this sum, taxes and social contributions paid are subtracted.
> Since every household has different compositions, the total income is divided by the number of "equivalent adults" using the modified OECD equivalence scale. This

scale assigns:

1. A weight of 1.0 to the first adult.
2. A weight of 0.5 to each additional household member aged 14 and over.
3. A weight of 0.3 to each child under 14.

The result is known as equivalized disposable income and it is assumed to be the same for each member of the household."

The income reference period usually covers a fixed 12-month span most of the times being the previous tax year. However, exceptions exist since for example in the UK, income data refer to the survey's current year. The indicator can be expressed in euros, national currencies or adjusted for purchasing power standard (PPS). The median disposable income is a reliable indicator of the purchasing power of the average person since as we said before its unaffected by outliers. Tracking both the median and mean though could help in highlighting income distribution trends. For example, if the average income rises while the median income remains stable, it can indicate that income gains are concentrated among the wealthiest, even if they represent only a small portion of the population.
(Eurostat, 2017)

According to Eurostat (2025f) filtering for Median Income:
Filtering for C.1:
Age: Total (every individual) [TOTAL]
Unit of Measure: Euro [EUR]
Sex: Total (both male, female) [T]
Income and living conditions indicator: Median equivalized net income [MED_E]
(Eurostat. (2025). Mean and median income by age and sex)

b) Consumption

Beyond just income, which as we said is the main financial tool for families to rely on covering daily living costs, what also matters is how people use that money. Many experts argue that observing both income and spending together offers a much more clear picture of someone's well-being than only focusing on income. That's mainly because while incomes can have fast changes, people often manage to keep their spending stable by tapping into savings or borrowing money. This stable nature of spending is one of the

reasons economist Milton Friedman developed the permanent income hypothesis, which suggests that individuals make financial decisions based on their long-term income expectations and not just what they're earning right now.

Lewis, Snape & Tonkin, 2014 (according to Eurostat, 2017) show that how much a household spends may actually matter more than how much it earns when it comes to overall well-being. As spending goes up, people tend to report greater life satisfaction and happiness. Higher spending doesn't seem to lead to more anxiety unlike income, which is closely tied to financial stress. The reason behind that may be that spending reflects the comforts people can afford while income is more about feeling secure.

The Stiglitz, Sen, and Fitoussi Commision, 2009 (according to Eurostat, 2017) emphasized that we should look at how income and spending are distributed across households and their interactions. If two households are spending the same amount but one has a higher income than the other, it is better off because of having more room to save or spend in the future if needed.

While GDP per capita is commonly used to measure a country's standard of living, it really only shows the total amount of goods and services a country produces. Actual Individual Consumption per capita reflects how people live by accounting also for differences in how countries fund services like education and healthcare. This is of high importance since in one country for example dental care might be paid from the government while other households must cover it by themselves. AIC makes international comparisons fairer and more accurate. It is also pointed out by the Commision that we need to keep in mind factors that don't show up as income but clearly impact well-being.

Even back in the 19th century, Engel's Law pointed out that as income increases, the percentage of income spent on food decreases even if people are spending more money overall. This idea extends to other basic household expenses as well.

Actual Individual Consumption (AIC)

Definition: AIC per capita, often expressed in purchasing power standards (PPS), refers to everything households actually consume even including services they receive from the governments such as healthcare or education.

Purchasing Power Parities (PPPs)

They are used to adjust for differences in price levels between countries. They can show

us how much services cost in different places. When national spending figures are converted using PPPs, they become a kind of artificial currency that evens out price differences between countries. This allows for more accurate cross-country comparisons since exchange rates alone can be misleading because they are affected by financial markets and trade volumes and not only prices. PPPs shows the real value of what households can afford in each country.
(Eurostat, 2017)

- **C.2 Gross Domestic Product Per Capita**

According to Eurostat "There are three ways of measuring GDP at market prices:

1. the production approach, as the sum of the values added by all activities which produce goods and services, plus taxes less subsidies on products;
2. the expenditure approach, as the total of all final expenditures made in either consuming the final output of the economy, or in adding to wealth, plus exports less imports of goods and services;
3. the income approach, as the total of all incomes earned in the process of producing goods and services plus taxes on production and imports less subsidies." (Eurostat, 2025k)

Filtering for C.2:
Unit of Measure: Chain Linked Volumes (2015) euro per capita
Chain Linked Volumes is an inflation adjusted indicators designed to not be affected by prince changes over time. Values are calculated in previous year's prices and then chain linked to a reference year (in our case 2015) [CLV15_EUR_HAB]
National accounts indicator: Actual Individual Consumption (AIS) [P41] (Eurostat, 2025c)

c) Material Condition: One more helpful tool in understanding people's well-being is the material deprivation indicator, which offers an additional view by focusing on people's actual living situations. This measure is of high importance since it highlights significant differences in living standards across countries.

Material Deprivation focuses on whether people lack basic services, experience financial difficulties or face barriers to living with dignity. As Atkinson et al., 2002 (according to Eurostat, 2017) pointed out, the ability to cope with unexpected expenses or the quality of ones home deeply affect their standard of living and overall quality of life.

The role of Housing

Housing quality if a significant part of material conditions. A home should be able to provide security privacy and personal space. When housing is dealing with having structural problems like damp walls or no working plumbing, or dealing with overcrowding, it can seriously harm physical and mental health. This is linked to domestic conflict, poor performance of children at school and limited ability in socializing since there is not enough space to host friends. This makes poor housing a key factor in well-being but in a negative aspect.

Housing indicators are used to get a full picture of people's well-being. These indicators include:

1. Structural problems (leaks, dampness, or rotting windows)
2. Overcrowding (not enough space for people inside the house)
3. Lack of basic amenities (toilets, showers)

Ewans et al., 2002 (according to Eurostat, 2017) supported that the overcrowding indicator is not just about comfort, it is also a health and well-being issue. Lack of space can increase the risk of mental strain and is especially stressful for children. At the same time, having too much space, which is known as under-occupation, is not necessarily ideal either since it is often tied with older adults living alone in large homes because their children have moved out. This may contribute to feelings of isolation.

(Eurostat, 2017)

- **C.3 – (In)ability to Make Ends Meet**

According to Eurostat (2017):

> "This indicator shows how people feel about their household's ability to cover essential expenses based on their total income. It comes from the EU-SILC survey question: Thinking about your household's total income, would you say your household is able to

make ends meet, that is, pay for its usual necessary expenses?

Respondents can choose from the following options:

1. With great difficulty (The percentages that came from this option were chosen for our promethea methodology)
2. With difficulty
3. With some difficulty
4. Fairly easily
5. Easily
6. Very easily

This is a measure, where individuals evaluate their financial situation considering all their sources of income and regular necessary expenses.

The indicator is expressed as the percentage of people selecting each level of difficulty or ease."

(Eurostat, 2017)

According to Eurostat (2025e) filtering for Inability to make ends meet:

C.3 Filters:

Subjective and non-monetary indicator: Households making ends meet with great difficulty [EM_GD]

Type of household: Total (total population of the household) [TOTAL]

Unit of measure: Percentage

Income situation in relation to the risk of poverty threshold: Total (Every individual, not separating between those below or above 60% of median equivalized income) [TOTAL]

- **C.4- Structural problems in the home**

According to Eurostat (2017):

"This indicator shows the percentage of people living in homes that deal with issues like:

1. Leaking roofs
2. Damp or moldy walls, floors, or foundations
3. Rotting window frames or floors"

According to Eurostat (2025i) filtering for Total population living in a dwelling with a leaking roof, damp walls, floors or foundation, or rot in window frames or floor:

C.4 Filters:
Type of household: Total (total population of the household) [TOTAL]
Sex: Total (both male and female) [T]
Age class: Total (every age) [TOTAL]
Unit of measure: Percentage
Income situation in relation to the risk of poverty threshold: Total (Every individual, not separating between those below or above 60% of median equivalized income) [TOTAL]

- **C.5- Overcrowding**

According to Eurostat (2017):

> "The indicator evaluates whether a household has enough rooms for its members based on:
>
> 1. One room for the household as a whole
> 2. One room per couple
> 3. One room per adult living alone
> 4. One room per pair of same-gender teens (12-17)
> 5. One room per single teen not included above
> 6. One room per pair of children under 12
>
> If a house has fewer rooms than this minimum, it is considered overcrowded. This means a person is considered to be living in an overcrowded household if their home doesn't have enough rooms to meet the basic space requirements for everyone living there. The overcrowding rate represents the percentage of the population affected by this lack of space". (Eurostat, 2017)

According to Eurostat (2025g) filtering for Overcrowding rate by age, sex and poverty status - total population

C.5 Filters:
Sex: Total (both male and female) [T]

Age class: Total (every age) [TOTAL]
Unit of measure: Percentage
Income situation in relation to the risk of poverty threshold: Total (Every individual, not separating between those below or above 60% of median equivalized income) [TOTAL]

2) Productive Or Other Main Activity

Work, whether it's paid or unpaid plays a central role in people's lives. Not only does it provide money for an individual but it also shapes their identity and provides valuable opportunities for social interaction. The amount of time one spends working can have significant impact on how fulfilled and happy they feel.

When someone's work provides learning, creativity and the achievement of their goals in addition to earning a decent income, it can boost a person's well-being. If the working environment though is full of discrimination, insecurity and low wages it can do the exact opposite, having a negative impact in a person's Quality of Life. Being unemployed also, has been shown to have serious negative effects on mental health.

The employment rate is the percentage of the population with a job that are in a working-age. It is a crucial indicator used to evaluate the health of the labor market. According to economists, a high employment rate, generally means more people working and contributing to GDP.

Unemployment on the other hand has significant negative effects as we mentioned. Research shows that people who are unemployed experience higher levels of stress, sadness, and anxiety and lower levels of joy and life satisfaction. It can also lead to crucial social problems, such as higher rates of divorce, suicide and substance abuse. Moreover, research from Abdallah, Stoll, and Eiffe, 2013 (according to Eurostat, 2017) shows that the connection between unemployment and a poor Quality of Life is direct. The longer someone remains unemployed the worse the effects. (Eurostat, 2017)

- **C.6 – Unemployment Rate**

Unemployment rate according to Eurostat (2017) "is a percentage of people unemployed within the active population of the same age group (15-74). In this category the persons that exist are those who were without work during the reference week, were currently available for work and were either actively seeking work in the past four weeks or had

already found a job to start within the next three months.
The labour force includes both employed and unemployed person." (Eurostat, 2017)

According to Eurostat (2025j) filtering for Unemployment rates by sex, age and educational attainment level (%):

C.6 Filters:
Sex: Total (both male and female) [T]
Age class: From 15 to 74 years [Y15-74]
Unit of measure: Percentage
International Standard Classification of Education: All ISCED 2011 levels (this includes all the education levels) [TOTAL]

3) Economic Security and Physical Safety

a) Economic Security: Economic security isn't just about how much income or wealth a person or household has it's also about how well they can deal with unexpected financial difficulties. That's the reason why indicators, such as how secure people feel, are great at capturing the level of economic stability or vulnerability. Other significant indicators are for example falling behind on rent or loan payments. Economic shocks can come in many forms. Some of them are job loss, declining health, aging-related expenses or financial crises in general. Shocks like these ones can push vulnerable households to their limits and beyond. (Eurostat, 2017)

- **C.7 – Population Unable to Cover Unexpected Financial Expenses**

This indicator shows the percentage of households that answered "no" to the following question in the EU-SILC survey:
"Can your household afford an unexpected required expense and pay for it using your own resources?"
Respondents can choose either yes or no.
In this concext according to Eurostat (2017), "own resources means that the household:

1. Does not rely on financial help from others
2. Is able to pay the full amount on time

3. Has not worsened their financial situation or gone further into dept as a result
4. Does not pay in instalments (or by taking a loan) expenses that previously used to pay in cash.

Required expenses could be different across countries. Some examples are surgery, funeral, major repair in the house or the vehicle an individual uses and more." (Eurostat, 2017)

According to Eurostat, 2025d filtering for Unexpected Expenses:
C.7 Filters:
Type of household: total (every member of the household) [TOTAL]
Unit of measure: Percentage
Income situation in relation to the risk of poverty threshold: Total (Every individual, not separating between those below or above 60% of median equivalized income) [TOTAL] (Eurostat, 2025d)

- **C.8 – Population in Arrears (Debt)**

According to Eurostat (2017) "the indicator shows the percentage of households that replied 'yes' (regardless of the frequency) to any of the following EU-SILC questions:
HS011: In the last twelve months, has the household been in arrears, i.e. has it been unable to pay on time due to financial difficulties for:
a) rent
b) mortgage repayment for the main dwelling

1. Yes, once
2. Yes, twice or more
3. No

HS021: In the last twelve months, has the household been in arrears, i.e has it been unable to pay on time due to financial difficulties for utility bills (heating, electricity, gas, water, etc.) for the main dwelling?

1. Yes, once
2. Yes, twice or more
3. No

HS031: In the last twelve months, has the household been in arrears on hire purchase instalments or other loan payments, i.e. has it been unable to pay on time due to financial difficulties?" (Eurostat, 2017)
*Please note that mortgage instalments for main dwelling are excluded

1. Yes, once
2. Yes, twice or more
3. No

major repair in the house or the vehicle an individual uses and more. (Eurostat, 2017)

According to Eurostat (2025a) filtering for Arrears (mortgage or rent, utility bills or hire purchase) from 2003 onwards:

C.8 Filters:
Type of household: total (every member of the household) [TOTAL]
Unit of measure: Percentage
Income situation in relation to the risk of poverty threshold: Total (Every individual, not separating between those below or above 60% of median equivalized income) [TOTAL]

b) Physical Safety: In addition to economic threats, people also face risks to their physical safety. Some examples could be crime, violence or accidents. Even if these risks may never actually happen, the feeling that they might, can seriously affect someone's sence of security and overall Quality of Life. This kind of safety means being protected from any kind of situation that puts someone's physical health or security in danger. People's perception of danger can sometimes have a bigger impact on their mental health than the actual threat itself. For example, homicide is a rare cause of death but it has a much stronger emotional impact on society than deaths caused by illness or accidents. Therefore, crimes that threaten personal safety don't just affect the direct victims but also leaves many others feeling unsafe, anxious or afraid.
To measure physical security, data like homicide rates can be used, which are the most consistent and comparable crime statistics available internationally. Other types of crime such as assaults or theft are less harmonised across countries. This makes them harder to compare.

Perception of crime doesn't always match reality. In areas where crime rates are low, residents might still feel unsafe. This can lead them for example to avoid going out at night, which may ironically reduce crime even further. Understanding how safe people feel is just as important as knowing the crime statistics when evaluating the true security of a community. (Eurostat, 2017)

- **C.9 – Perception of crime, violence or vandalism in the living area**

According to Eurostat (2017):

> "This indicator concerns the percentage of household that replied yes to the following question:
> Do you have any of the following problems related to the place where you live --- crime, violence and vandalism in the local area?
> The goal is to assess whether the respondent feels 'crime, violence or vandalism' to be a problem for the household. Crime is to be defined as a deviant behaviour that violates prevailing norms, specifically, cultural standards prescribing how humans ought to behave normally". (Eurostat, 2017)

According to Eurostat (2025b) filtering for Crime:
C.9 Filters:
Type of household: total (every member of the household) [TOTAL]
Unit of measure: Percentage
Income situation in relation to the risk of poverty threshold: Total (Every individual, not separating between those below or above 60% of median equivalized income) [TOTAL]

4) Natural And Living Environment

Pollution: According to Eurostat, 2017 the fact that environmental indicators are extremely important in other contexts makes them relatively abundant. Nevertheless, their contribution in providing valuable information about quality of life especially when combining them with self-reported quality of one's environment, is certain. In this research done by Eurostat, 2017 its mentioned that its highly proven that environmental problems

and pollution have high association with health problems and well-being with one example being air pollution. Air pollution can have both immediate and long-term effects on health. In the short term, it may lead to respiratory infections like bronchitis or pneumonia, and worsen existing conditions such as asthma and emphysema. Over time, exposure to polluted air can contribute to serious illnesses affecting the lungs and heart, ultimately reducing life expectancy. Eurostat, 2017 also states that beyond its health impact, air pollution also places a financial burden on the economy. These costs can be direct, such as increased spending on medications and healthcare services, or indirect such as reduced productivity due to illness-related absences. The loss of life years caused by pollution also represents a decline in human capital. For instance, its also mentioned in Eurostat (2017) that in the United Kingdom, air pollution is estimated to shorten average life expectancy by about six months, with related healthcare and economic costs reaching up to £19 billion annually. Additionally, pollution harms the environment, damaging ecosystems and plant life, and contributed significantly to the formation of smog. Due to its relevance, one measurable indicator of air pollution was selected as part of the well-being framework. (Eurostat, 2017)

- **C.10 Perception of pollution, grime or other environmental problems in the living area**

Definition: According to Eurostat (2017) "this indicator has been calculated as the percentage of the households that have answered "Yes" to the following question: *Do you have any of the following problems related to the place where you live: pollution, grime or other environmental problems in the local area such as: smoke, dust, unpleasant smells or polluted water?*

Answer categories: yes, no
The objective is to assess whether the people questioned feel pollution, grime or other environmental problems in the living area." (Eurostat, 2017)

According to Eurostat (2025h) filtering for Pollution, grime or other environmental problems:
C.10 Filters:
Type of household: total (every member of the household) [TOTAL]
Unit of measure: Percentage

Income situation in relation to the risk of poverty threshold: Total (Every individual, not separating between those below or above 60% of median equivalized income) [TOTAL] (Eurostat, 2025h)

2.7 Quality of Life Rankings

According to Numbeo, 2022 Quality of Life rankings for 2022 for our European Union countries selected is as follows:

1. Denmark
2. Netherlands
3. Sweden
4. Luxembourg
5. Spain
6. Portugal
7. Slovakia
8. Hungary
9. Romania
10. Greece

OECD (2024) ranks European Union countries selected as follows:

1. Sweden
2. Netherlands
3. Denmark
4. Luxembourg
5. Spain
6. Slovakia
7. Hungary
8. Portugal
9. Greece

Romania is missing from the data.

2.8 Research Aim and Questions

The aim of this research is to assess and compare the Quality of Life across ten selected European union countries using the PROMETHEE II methodology. A complete ranking of these countries based on specific indicators chosen derived from Eurostat will occur after a python implementation of the PROMETHEE II methodology. Using Visual PROMETHEE we will also generate preference scores and overall rankings for each country as-well. By doing this we will compare outputs that came from our python implementation with the ones from Visual PROMETHEE.

Research Questions

1. **For the yearly scenario, which EU countries rank the highest, lowest? Give the rankings using all preference functions.**
2. **For the average scenario, which EU countries rank the highest, lowest? Give the rankings using all preference functions**
3. **Does the change in Preference functions affect the final ranking? What function gives the best first and worst last place?**
4. **Using any function preferred, does using the weights derived from the questionnaire give significantly different rankings compared to the weights given from the researcher? Give examples**
5. **Do the highest-lowest performance countries change in average, yearly scenarios?**
6. **What year was the best and worst for every country? Is there any particular year that seems for every country was a downfall?**
7. **To what extent did covid19 period affect Quality of Life?**
8. **Which countries showed the most improvement or decline?**
9. **GDP used to be the one of the most significant indicators. To what extent, countries with similar GDP profile rank the same?**
10. **Using any function preferred, are the results from python, the same with the ones from visual PROMETHEE?**
11. **Are there countries that overall perform extremely low at some indicators but excel in others?**
12. **Which indicators have the most impact?**

13. **Are there criteria that act the same and correlate?**
14. **What is the effect of removing one criterion at a time on the final ranking?**
15. **Which countries are the best, worst across all indicators?**

3. Methodology Overview

The implementation of the PROMETHEE II methodology for evaluating Quality of Life across selected European Union countries followed a structured, multi-step process, combining data preprocessing, custom Python programming, and validation through Visual PROMETHEE software. This study uses the methodology of PROMETHEE as seen in the work of Papathanasiou and Ploskas (2018), in "Multiple Criteria Decision Aid" expanding it in comparing the well-being of countries in the European Union. While previous literature has applies MCDA methods to evaluate Quality of Life at urban, national or individual levels, this research aims to do something more specific. Our research aim is to compare quality of life in 10 countries of the European Union, using 10 different well-being indicators. The data derived from Eurostat and every Preference Function for the PROMETHEE methodology will be used. Different weighting scenarios are also introduced with both the researchers view and the participants of a questionnaire about the significancy of the criteria in the overall well-being. Finally, Visual PROMETHEE was used for cross-validation and enhanced visualization of the results. Below, is a detailed breakdown of the extension of the methodology for our research:

3.1 Data Collection, Missing Data

All relevant data were sourced from Eurostat, focusing on indicators closely related to Quality of Life.

To collect and prepare the data we wanted to use for analysis, a function was developed to automate the retrieval and preprocessing of several datasets from Eurosta. This function takes four inputs: the dataset code, a set of filters, the desired name for the value column, and a list of selected countries. Once the data are retrieved from Eurostat, they are filtered to the conditions we choose. The function also extracts data specifically for the years 2014 to 2023, restructures the dataset from a wide to a long format using pandas.melt(), and standardizes country names. This function ensures that only the selected EU countries and appropriate indicators are retained, while also making the data in a format suitable for further analysis. Using this function it is also quite easy to process for any country we want for further analysis and examples. To ensure the consistency of the dataset, since in some criteria missing values existed, they were addressed through mean imputation. Instead of removing the incomplete records, the average value for each criterion within each country was calculated so that the missing entries were filled

accordingly. Doing this we maintained the datasets structure while also avoiding data loss, in this 10-year range.

3.2 Selection of Countries, Criteria

For this study we needed ten European Union countries. The selection was done with the goal of having a diverse sample in terms of economic development and Quality of Life factors. The countries include:

Country	ISO Code
Denmark	DK
Greece	EL
Spain	ES
Luxembourg	LU
Netherlands	NL
Portugal	PT
Romania	RO
Slovakia	SK
Sweden	SE
Hungary	HU

The selection was aiming to include countries in all the spectrum of socioeconomic performance. Denmark, Luxembourg and Sweden are countries with high median income and GDP per capita. This is connected with strong well-being and high living standards. Spain, Portugal, Greece and Slovakia represent medium income economies, each of them dealing with social and economic pressure. Hungary, Romania and Slovakia offer insight into lower-income countries, allowing our analysis to observe how different levels of economies impact on Quality of Life. The versatility also these countries have across the criteria chosen is also a strong reason of selection. While some have strong performance in certain areas, they rank poorly in others. For example, Greece has low income and high unemployment rates but moderate levels in the housing deprivation indicator. Hungary may cope with housing conditions and overcrowding but has the lowest unemployment rate. The selection of criteria was seen also in the literature part before. Their significancy in Quality of Life was also mentioned. There were also other criteria that would be of

much help in describing Quality of Life but some of them were missing several annual reports in Eurostat. So, the ten criteria were chosen also for their consistency in reporting across the selected EU countries. They represent various dimensions of well-being such as income, employment, housing conditions and environmental concerns. These indicators in our python script gathered together are:

1. C.1 Median income
2. C.2 GDP per capita
3. C.3 Ends meet
4. C.4 H. Deprivation
5. C.5 H. Overcrowding
6. C.6 Unemployment
7. C.7 Unexpected Expenses
8. C.8 Arrears
9. C.9 Crime
10. C.10 Pollution

After having our indicators selected, it is necessary to define whether a criterion should either be maximized or minimized. Criteria that have positive correlation with Quality of Life, such as Median Income and GDP per Capita, were defined as "to be maximized". On the other hand, indicators like Unemployment, Pollution, or Arrears were defined as "to be minimized" since they have negative relationship with well-being.

3.3 Dataset Creation

In our script 2 scenarios will be covered. One is based on all entries of every country across the years. This one we call it as the yearly scenario. It will perform PROMETHEE II for every country in every year in the range of 2014-2023. This means that every country will compare even with itself so that in the end a complete ranking will emerge. This ranking will be like:
Greece 2014 < Hungary 2014 < Denmark 2021 etc.
For the second scenario, an aggregated dataset was constructed to support an average-based analysis. For each country, the mean value of each criterion over the 10 year period was computed, resulting in a single value per indicator. This aggregation will help in

having to deal with less data than the yearly method so in some examples we want to visualize, outputs will be easier.

3.4 Preference Function and Weights

In our python script the user is allowed to select the preference function that will be applied across all criteria from a predefined list. This customization helps in observing how all these different approaches will either give same or different rankings and which one gives the highest or lowest. The weights of each criterion represent the importance the researcher thinks each indicator has for Quality of Life. In our python script 2 options will be used:

1. Manually selected weights, based on my judgment
2. Weights, derived from a questionnaire we made distributed as a part of the study

In this survey 100 university students and graduates were asked to measure each criterion in a scale 1-5 based on the importance they think each indicator has on well-being. Normalization was later done based on the data to create a weighted profile used in the analysis. A total of 100 university students and graduates were asked to participate in a questionnaire designed to observe the significance of our 10 criteria based on their opinion. Each participant was asked to rank the importance of each criteria on a scale from 1 to 5, with 1 indicating low ranking and 5 high. This ranking aimed to capture the perspective of the participants on how important each of the indicators is on the criteria and then using their opinions weights were created. To create weights based on the answers of the participants, a Python script was performed. After loading the Excel file containing the participants answers, the average score for each criterion was calculated. The average scores were then normalized by dividing each by the total sum of averages. This process resulted in a set of normalized weights that show how important each indicator was considered by the participants of the questionnaire. The weights that occur from the normalization are:
w = [0,099251, 0,097111, 0,120385, 0,101926, 0,073569, 0,099518, 0,113162, 0,111557, 0,094703, 0,088818]

3.5 Definition of the Unical Function

PROMETHEE II is about how much an indicator is preferred over another. For this reason preference functions are crucial. They introduce as we said before different forms of computing preference across criteria. Unical function was developed, to help us use every preference function depending what the user wants to observe. While later computing the preference matrix, it will be used and will enable the user in having flexibility to the decision making.
Unical Parameters:

1. diff_value: The difference between two alternatives values for a given criterion.
2. q: The indifference threshold
3. p: The preference threshold.
4. f: A string indicating the type of preference function to be applied ('u' , 'vs', 'g', etc.)

If invalid preference function type is entered, the function rases a ValueError, helping prevent incorrect analysis.

3.6 Difference Matrix Computation

The function compute_differences(x, c) was created to calculate the pairwise comparisons of every action for each criterion. The purpose of this function is to generate difference matrices for each criterion.
Parameters

1. X: A 2D NumPy array where each row represents a country and each column represents an indicator. This matrix can contain either the aggregated or yearly data.
2. C: A binary list that shows the direction of preference for each criterion. The value can either be 1 when the indicator is to be maximized, or 0 when it is to be minimized.

Each difference matrix computed is stored in a dictionary with the criterion name as the key. This means in total that the output is a dictionary of NumPy matrices, one for each

indicator where each matrix shows the pairwise differences of each country in a given criterion.

3.7 Threshold Calculation

As we said, several preference functions need the use of thresholds. These thresholds help define indifference, partial preference and absolute preference based on the correlation between alternatives. To automate the estimation of these thresholds p, q, s, the compute thresholds function was developed. Its purpose is to compute the thresholds p,q for each criterion. To compute them, the standard deviation of each criterion was used. A quarter of the standard deviation is used for calculating the indifference threshold while three-quarters for the preference threshold.
The function returns a NumPy array that contains both p,q values structures as two lists.

3.8 Preference Matrix

Once the pairwise differences between the countries have been calculated, the next step is to convert these differences into preference degrees to observe how each country is preferred over the others. Compute preference matrix function plays a crucial role in this. Its purpose is to take the difference matrices computed for each indicator and change them in preference matrices that will show the strength of preference of one country over another. The values inside the preference matrices range between 0(no preference) and 1(strong preference). In the case of the usual preference function only 0 and 1 will be inside the matrix but when using other preference functions, like the level for example, the values inside the matrix can have more values, in this case 0.5 also. It depends on the preference function chosen from the user.
Parameters

1. Diff_matrices: A dictionary that contains the pairwise difference matrices for each criterion. This is previously computed using the compute differences function.
2. q: An array with indifference thresholds inside (one per criterion)
3. p: An array with preference thresholds inside (one per criterion)
4. f: A list showing the preference function to be used for each criterion

3.9 Flow Calculation

Once the preference matrices have been calculated, the preference flows can be computed. These flows show how each country compares to all others across each indicator. Compute flows function was designed to calculate the positive, negative flows and netflows. These flows are vital in determining how favorable a country is compared to others.

Parameters

1. Preference matrices: A dictionary where each key is a criterion and each value is a NumPy matrix that represents the preference degrees among the countries for a specific criterion.

The function returns a dictionary containing all three types of flows for each criterion.

3.10 Weighted and global flows

After the calculation of the net flows for each country on each criterion, the final step is to combine these flows to find the overall ranking of each alternative. Compute weighted flows function was designed to aggregate the net flows from each criterion into a global net flow for each country. For each criterion, the function multiplies the net flow vector by its corresponding weight previously assigned. All weighted flows are then summed across all criteria to produce a single global net flow that shows the final performance score for each country.

Parameters

1. Unicriterion flows: A dictionary containing the net flows, previously calculated from compute flows function
2. Weights: A NumPy array of normalized values summing to 1, representing the importance of each criterion.

The function returns a dictionary of weighted net flow vectors for each indicator and a NumPy array with the overall score for each country inside.

Exporting Results to Excel

A save_to_excel function was developed to export significant results to excel. For the

average scenario, there are details about difference, preference matrices, positive, negative flows and net flows for every indicator. For the yearly scenario, this is only done for 1 selected indicator for cross validation that the algorithm works correctly. For both of them, the python script outputs the final global net flows and rankings.

3.11 Visualization

To interpret the results, a save_to_excel function was developed to export significant results to excel. For the average scenario, there are details about difference, preference matrices, positive, negative flows and net flows for every indicator. For the yearly scenario, this is only done for 1 selected indicator for cross validation that the algorithm works correctly. For both of them, the python script outputs the final global net flows and rankings. Two more visualization methods were implemented.
The directed graph function generates a directed flowchart-style diagram that ranks the countries based on their global net flows. Using the Graphviz library this graph emerges and saved as a pdf file kept for extra study. Works great for both average and yearly scenarios.
Parameters

1. Global net flow: The final global net flow values for each country
2. Country names: The list of country names
3. Title: A string used for the title and output file name

The scatterplot function uses Matplotlib to generate a 2D plot of countries on the x axis and their net flow scores on the y axis.
Parameters

1. Global net flow: The final global net flow values for each country
2. Country names: The list of country names
3. Title: A string used for the title of the plot

These plots purpose is to observe the distribution of the net flow scores to spot patterns. In the average scenario the interpretation is easier due to less data.

3.12 Visual PROMETHEE Cross-Validation

To ensure the reliability and accuracy of the results we got from our python implementation, Visual PROMETHEE was later used for cross-validation. Due to less data on the average-based scenario, this one was used. Visual PROMETHEE includes visual tools that enhance the interpretability of the results. Some of them used:

1. The Ranking Table, that provides the net flows
2. The Ranbow Graph, a visual showing how each country performs across indicators
3. The Gaia Plane, a 2D projection that gives extra information about criteria and actions
4. The Walking Weight, which demonstrates how changing the values of the weights in an indicator affects rankings

4. Results

In this section, the results obtained through our methodology are presented. These include outputs from our Python script, the Excel datasets and the Visual PROMETHEE software. These outputs are the foundation of our research and seem vital in comparing Quality of Life across the European Union countries we chose to observe.
The results include rankings net flow values and visualizations. This is done for 2 different weights scenario by using all 6 preference functions as well.

Methodology for Average Scenario using Usual Preference Function for Median Income Indicator:
1) Difference Matrix

Country	DK	EL	ES	HU	LU	NL	PT	RO	SE	SK
DK	0	22156.5	15365.6	24743.1	-7781.8	5489.3	20641.3	26705.4	4914.2	22636.3
EL	-22156.5	0	-6790.9	2586.6	-29938.3	-16667.2	-1515.2	4548.9	-17242.3	479.8
ES	-15365.6	6790.9	0	9377.5	-23147.4	-9876.3	5275.7	11339.8	-10451.4	7270.7
HU	-24743.1	-2586.6	-9377.5	0	-32524.9	-19253.8	-4101.8	1962.3	-19828.9	-2106.8
LU	7781.8	29938.3	23147.4	32524.9	0	13271.1	28423.1	34487.2	12696	30418.1
NL	-5489.3	16667.2	9876.3	19253.8	-13271.1	0	15152	21216.1	-575.1	17147
PT	-20641.3	1515.2	-5275.7	4101.8	-28423.1	-15152	0	6064.1	-15727.1	1995
RO	-26705.4	-4548.9	-11339.8	-1962.3	-34487.2	-21216.1	-6064.1	0	-21791.2	-4069.1
SE	-4914.2	17242.3	10451.4	19828.9	-12696	575.1	15727.1	21791.2	0	17722.1
SK	-22636.3	-479.8	-7270.7	2106.8	-30418.1	-17147	-1995	4069.1	-17722.1	0

Figure 7: Difference Matrix

2) Preference Matrix

Country	DK	EL	ES	HU	LU	NL	PT	RO	SE	SK
DK	0	1	1	1	0	1	1	1	1	1
EL	0	0	0	1	0	0	0	1	0	1
ES	0	1	0	1	0	0	1	1	0	1
HU	0	0	0	0	0	0	0	1	0	0
LU	1	1	1	1	0	1	1	1	1	1
NL	0	1	1	1	0	0	1	1	0	1
PT	0	1	0	1	0	0	0	1	0	1
RO	0	0	0	0	0	0	0	0	0	0
SE	0	1	1	1	0	1	1	1	0	1
SK	0	0	0	1	0	0	0	1	0	0

Figure 8: Preference Matrix

3) Net flows

Country	Positive_Flow (Φ^+)	Negative_Flow (Φ^-)	Net_Flow (Φ)	Weighted_NetFlow
LU	1	0	1	0.15
DK	0.888888889	0.111111111	0.777777778	0.116666667
SE	0.777777778	0.222222222	0.555555556	0.083333333
NL	0.666666667	0.333333333	0.333333333	0.05
ES	0.555555556	0.444444444	0.111111111	0.016666667
PT	0.444444444	0.555555556	-0.111111111	-0.016666667
EL	0.333333333	0.666666667	-0.333333333	-0.05
SK	0.222222222	0.777777778	-0.555555556	-0.083333333
HU	0.111111111	0.888888889	-0.777777778	-0.116666667
RO	0	1	-1	-0.15

Figure 9: Positive, Negative Flows and Net Flows

4) Countries Performance Across all Indicators for Usual Preference Function

Φ Preference Flows

	GDP per	Median	Unemploym	Ends_Meet	Unexpected	Arrears	Overcrowdin	Housing_Con	Crime	Pollution
Denmark	0,7778	0,7778	0,3333	0,5556	0,3333	0,3333	0,3333	-0,1111	0,7778	0,7778
Greece	-0,3333	-0,3333	-1,0000	-1,0000	-1,0000	-1,0000	-0,5556	0,3333	-0,7778	-1,0000
Spain	0,1111	0,1111	-0,7778	-0,1111	-0,3333	-0,3333	0,7778	-0,5556	-0,1111	0,3333
Luxemburg	1,0000	1,0000	0,1111	0,3333	0,5556	0,5556	0,5556	-0,3333	-0,3333	-0,7778
Netherlands	0,3333	0,3333	0,7778	1,0000	1,0000	1,0000	1,0000	0,1111	-1,0000	-0,5556
Portugal	-0,1111	-0,1111	-0,5556	-0,5556	-0,1111	0,1111	0,1111	-1,0000	0,3333	-0,1111
Romania	-1,0000	-1,0000	0,5556	-0,7778	-0,7778	-0,7778	-1,0000	0,5556	0,1111	-0,3333
Slovakia	-0,5556	-0,5556	-0,3333	0,1111	0,1111	-0,1111	-0,7778	1,0000	1,0000	0,5556
Sweden	0,5556	0,5556	-0,1111	0,7778	0,7778	0,7778	-0,1111	0,7778	-0,5556	1,0000
Hungary	-0,7778	-0,7778	1,0000	-0,3333	-0,5556	-0,5556	-0,3333	-0,7778	0,5556	0,1111

Figure 10: Preference Flows

4.1 Outputs for Every Scenario, Preference Function

a) Usual Preference Function:

Average Scenario:

Table 1: Average Scenario, Usual Function Rankings

Country	Global_NetFlow	Rank
NL	0.5	1
DK	0.477777778	2
SE	0.424444444	3
LU	0.371111111	4
SK	-0.02	5
ES	-0.182222222	6
PT	-0.228888889	7
HU	-0.237777778	8
RO	-0.431111111	9
EL	-0.673333333	10

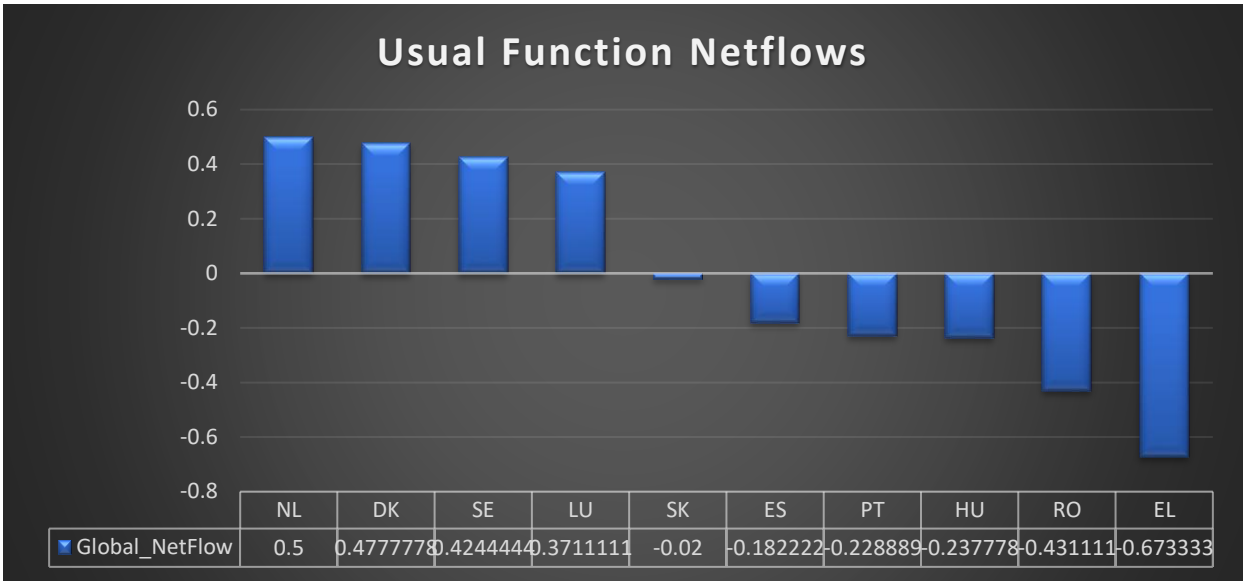


Figure 11: Average Scenario, Usual Function Rankings

Table 2: Yearly Scenario, Usual Function Rankings

Country	Year	Global_NetFlow	Rank	--------	Country	Year	Global_NetFlow	Rank
NL	2022	0.573636364	1		EL	2019	-0.568989899	91
DK	2021	0.573030303	2		EL	2018	-0.572121212	92
NL	2023	0.557777778	3		RO	2015	-0.596565657	93
DK	2022	0.54030303	4		EL	2017	-0.621515152	94
LU	2022	0.537070707	5		HU	2015	-0.630909091	95
NL	2021	0.529393939	6		EL	2016	-0.636565657	96
NL	2020	0.513535354	7		RO	2014	-0.653131313	97
LU	2019	0.497373737	8		EL	2015	-0.658484848	98
LU	2023	0.490707071	9		EL	2014	-0.662222222	99
DK	2019	0.488080808	10		HU	2014	-0.757575758	100

Average Scenario using weights from the questionnaire:

Table 3: Average Scenario, Usual Function Rankings ($W_{questionnaire}$)

Country	Global_NetFlow	Rank
DK	0.483622	1
SE	0.473753333	2
NL	0.428808667	3
LU	0.278639778	4
SK	0.060043778	5
ES	-0.124189111	6
PT	-0.216217333	7
HU	-0.259497556	8
RO	-0.445515333	9
EL	-0.679448222	10

Yearly Scenario using weights from the Questionaire:

Table 4: Yearly Scenario, Usual Function Rankings ($W_{questionnaire}$)

Country	Year	Global_NetFlow	Rank	----------	Country	Year	Global_NetFlow	Rank
DK	2021	0.561724879	1		EL	2020	-0.57146098	91
DK	2022	0.515864939	2		EL	2018	-0.577227606	92
NL	2022	0.486218051	3		RO	2015	-0.601121475	93
SE	2017	0.474236889	4		EL	2017	-0.619220485	94
DK	2019	0.469472687	5		EL	2016	-0.626800232	95
NL	2023	0.466734778	6		HU	2015	-0.639336364	96
LU	2023	0.463611768	7		EL	2015	-0.647983121	97
NL	2021	0.456620364	8		EL	2014	-0.663083374	98
DK	2023	0.450189697	9		RO	2014	-0.664097394	99
DK	2017	0.44476899	10		HU	2014	-0.762107556	100

b) U-shape Preference Function:

Average Scenario:

Table 5: Average Scenario, U-shape Function Rankings

Country	Global_NetFlow	Rank
DK	0.503333333	1
LU	0.482222222	2
SE	0.365555556	3
NL	0.362222222	4
SK	0.01	5
ES	-0.152222222	6
PT	-0.197777778	7
HU	-0.322222222	8
RO	-0.391111111	9
EL	-0.66	10

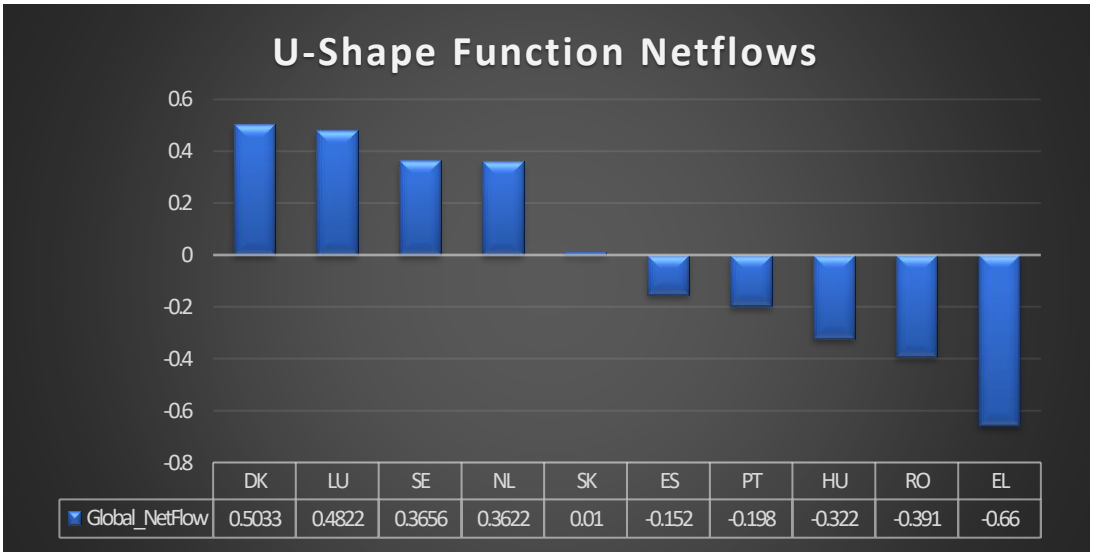


Figure 12: Average Scenario, U-shape Function Rankings

Yearly Scenario:

Table 6: Yearly Scenario, U-shape Function Rankings

Country	Year	Global_NetFlow	Rank	--------	Country	Year	Global_NetFlow	Rank
DK	2022	0.525050505	1		EL	2019	-0.578686869	91
DK	2021	0.518484848	2		EL	2018	-0.581414141	92
DK	2023	0.510606061	3		HU	2015	-0.588484848	93
LU	2022	0.501515152	4		EL	2020	-0.590606061	94
DK	2019	0.481919192	5		RO	2014	-0.592929293	95
NL	2022	0.473030303	6		EL	2017	-0.618080808	96
LU	2023	0.471111111	7		EL	2016	-0.621212121	97
NL	2023	0.465454545	8		EL	2015	-0.638989899	98
LU	2019	0.459191919	9		EL	2014	-0.663939394	99
NL	2021	0.451717172	10		HU	2014	-0.693131313	100

c) V-shape Preference Function

Average Scenario:

Table 7: Average Scenario, V-shape Function Rankings

Country	Global_NetFlow	Rank
DK	0.487154317	1
LU	0.453546604	2
SE	0.407254911	3
NL	0.370613875	4
SK	0.007304184	5
PT	-0.185157424	6
ES	-0.18832178	7
HU	-0.289751841	8
RO	-0.381400005	9
EL	-0.681242841	10

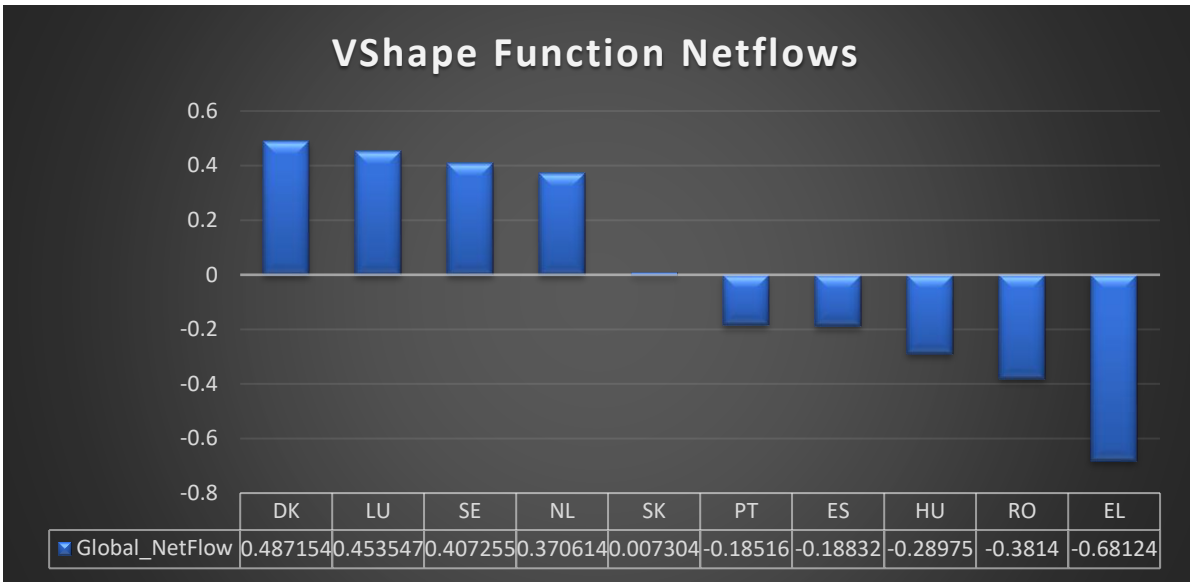


Figure 13: Average Scenario, V-shape Function Rankings

Yearly Scenario:

Table 8: Yearly Scenario, V-shape Function Rankings

Country	Year	Global_NetFlow	Rank	----------	Country	Year	Global_NetFlow	Rank
DK	2022	0.49512555	1		RO	2014	-0.559164737	91
DK	2021	0.490192934	2		EL	2021	-0.566099199	92
LU	2023	0.480513695	3		EL	2018	-0.591853385	93
LU	2022	0.471004142	4		EL	2019	-0.59625286	94
DK	2023	0.469842305	5		EL	2020	-0.598573516	95
DK	2019	0.449374352	6		EL	2016	-0.623302571	96
LU	2019	0.446454411	7		EL	2017	-0.624140344	97
NL	2022	0.431790304	8		EL	2015	-0.641216117	98
NL	2023	0.419688214	9		HU	2014	-0.648074023	99
DK	2017	0.413914038	10		EL	2014	-0.664913486	100

d) Level Function:

Average Scenario:

Table 9: Average Scenario, Level Function Rankings

Country	Global_NetFlow	Rank
DK	0.471666667	1
LU	0.440555556	2
SE	0.403333333	3
NL	0.333333333	4
SK	0.048333333	5
ES	-0.177777778	6
PT	-0.178333333	7
HU	-0.291666667	8
RO	-0.371111111	9
EL	-0.678333333	10

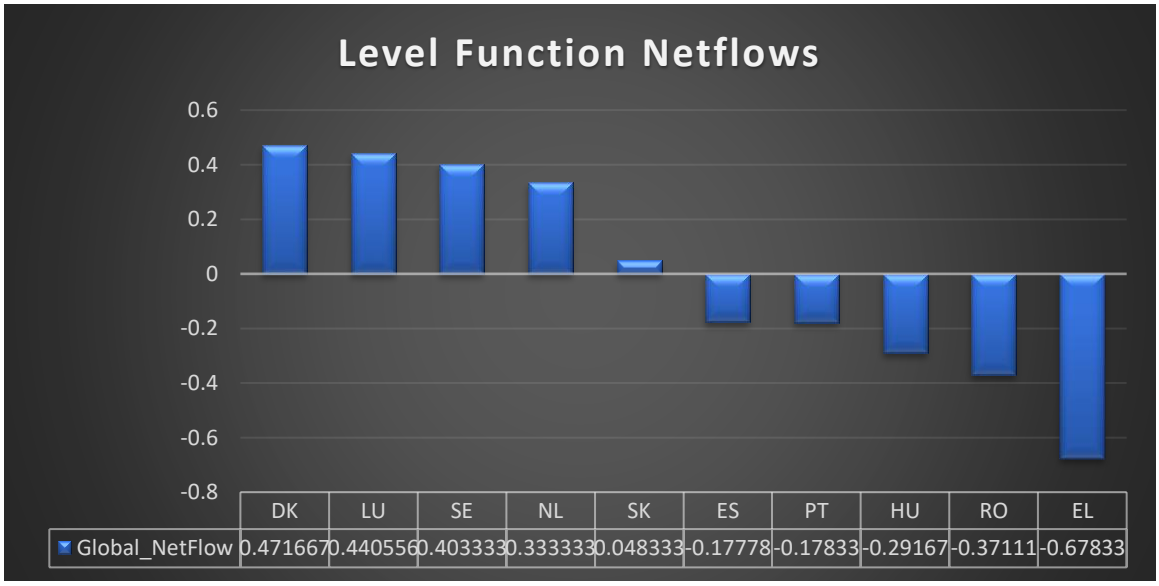


Figure 14: Average Scenario, Level Function Rankings

Yearly Scenario:

Table 10: Yearly Scenario, Level Function Rankings

Country	Year	Global_NetFlow	Rank	---	Country	Year	Global_NetFlow	Rank
DK	2022	0.464343434	1		EL	2023	-0.556666667	91
LU	2023	0.463232323	2		EL	2021	-0.571060606	92
DK	2021	0.461767677	3		EL	2018	-0.590707071	93
DK	2023	0.454343434	4		EL	2019	-0.599343434	94
LU	2022	0.451717172	5		EL	2020	-0.605656566	95
DK	2019	0.433787879	6		HU	2014	-0.615050505	96
LU	2019	0.411111111	7		EL	2016	-0.615808081	97
LU	2021	0.400707071	8		EL	2017	-0.618585859	98
SE	2023	0.398333333	9		EL	2015	-0.630151515	99
DK	2017	0.393080808	10		EL	2014	-0.660808081	100

e) V-shape with indifference (linear):

Average Scenario:

Table 11: Average Scenario, V-shape with Indifference (Linear) Function Rankings

Country	Global_NetFlow	Rank
DK	0.491353342	1
LU	0.463008918	2
SE	0.416334393	3
NL	0.331562564	4
SK	0.007643676	5
PT	-0.167133898	6
ES	-0.203646561	7
HU	-0.289103709	8
RO	-0.356437078	9
EL	-0.693581646	10

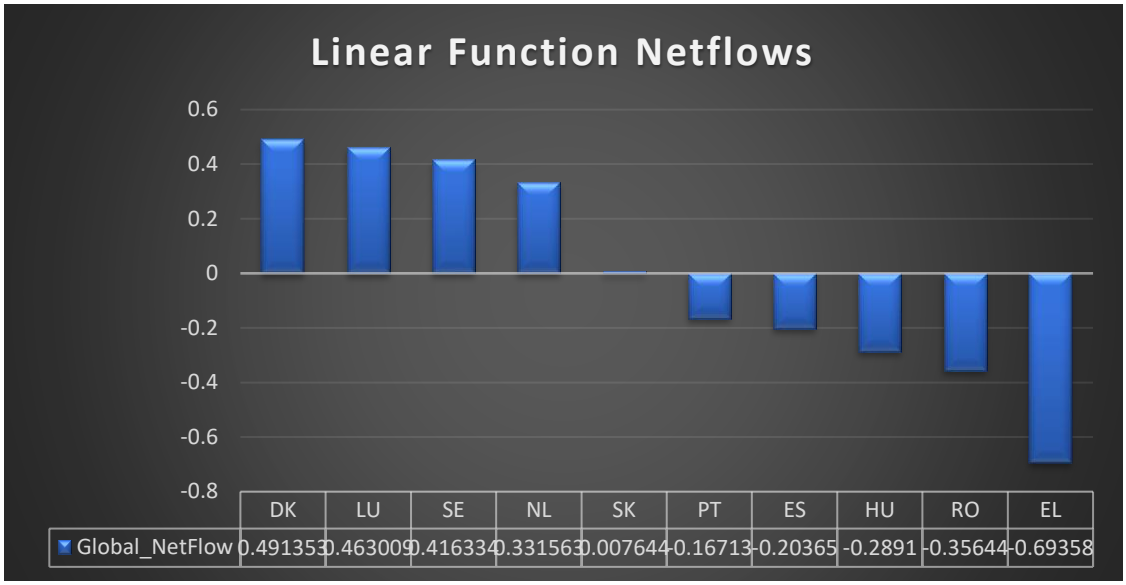


Figure 15: Average Scenario, V-shape with Indifference (Linear) Function Rankings

Yearly Scenario:

Table 12: Yearly Scenario, V-shape with Indifference (Linear) Function Rankings

Country	Year	Global_NetFlow	Rank	--------	Country	Year	Global_NetFlow	Rank
LU	2023	0.481457488	1		EL	2023	-0.565312104	91
DK	2022	0.471125076	2		EL	2021	-0.579013229	92
DK	2021	0.465529524	3		EL	2018	-0.599546288	93
DK	2023	0.461051142	4		HU	2014	-0.606194235	94
LU	2022	0.444655762	5		EL	2019	-0.607739764	95
DK	2019	0.436399968	6		EL	2020	-0.609123311	96
LU	2019	0.423105783	7		EL	2016	-0.620254208	97
DK	2020	0.410459302	8		EL	2017	-0.627082862	98
LU	2021	0.407904888	9		EL	2015	-0.636504734	99
DK	2017	0.4067714	10		EL	2014	-0.668749477	100

f) Gaussian Preference Function:

Average Scenario:

Table 13: Average Scenario, Gaussian Function Rankings

Country	Global_NetFlow	Rank
DK	0.487154317	1
LU	0.453546604	2
SE	0.407254911	3
NL	0.370613875	4
SK	0.007304184	5
PT	-0.185157424	6
ES	-0.18832178	7
HU	-0.289751841	8
RO	-0.381400005	9
EL	-0.681242841	10

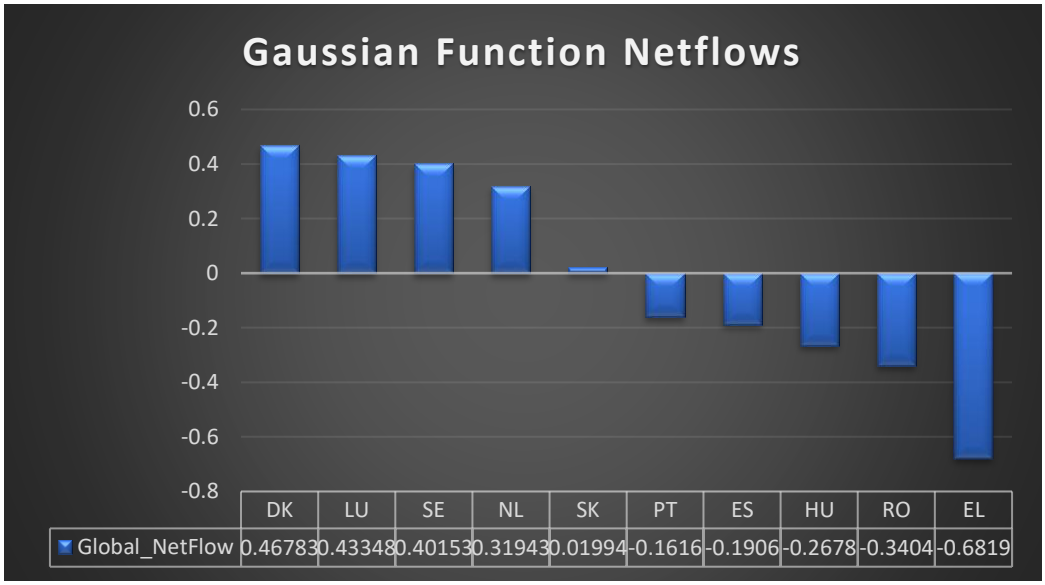


Figure 16: Average Scenario, Gaussian Function Rankings

Yearly Scenario:

Table 14: Yearly Scenario, Gaussian Function Rankings

Country	Year	Global_NetFlow	Rank	-------	Country	Year	Global_NetFlow	Rank
DK	2022	0.49512555	1		RO	2014	-0.559164737	91
DK	2021	0.490192934	2		EL	2021	-0.566099199	92
LU	2023	0.480513695	3		EL	2018	-0.591853385	93
LU	2022	0.471004142	4		EL	2019	-0.59625286	94
DK	2023	0.469842305	5		EL	2020	-0.598573516	95
DK	2019	0.449374352	6		EL	2016	-0.623302571	96
LU	2019	0.446454411	7		EL	2017	-0.624140344	97
NL	2022	0.431790304	8		EL	2015	-0.641216117	98
NL	2023	0.419688214	9		HU	2014	-0.648074023	99
DK	2017	0.413914038	10		EL	2014	-0.664913486	100

Using the Usual Preference Function:

4.2 Performance of Countries

Denmarks Performance Over the Years:

Table 15: Denmark 2014-2023, Usual Function

Country	Year	Global_NetFlow	Rank
DK	2014	0.216565657	38
DK	2015	0.349292929	28
DK	2016	0.404848485	17
DK	2017	0.436262626	13
DK	2018	0.38030303	24
DK	2019	0.488080808	10
DK	2020	0.402929293	18
DK	2021	0.573030303	2
DK	2022	0.54030303	4
DK	2023	0.461111111	11

Its visualization:

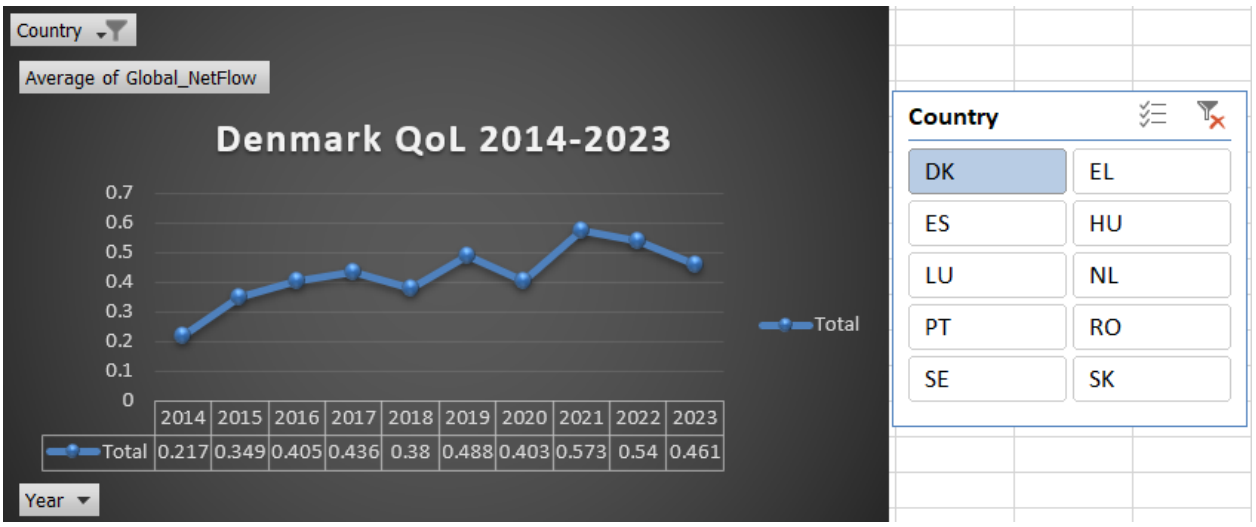


Figure 17: Denmark 2014-2023 Usual Function

Its indicators profile for the average scenario using Usual preference function:

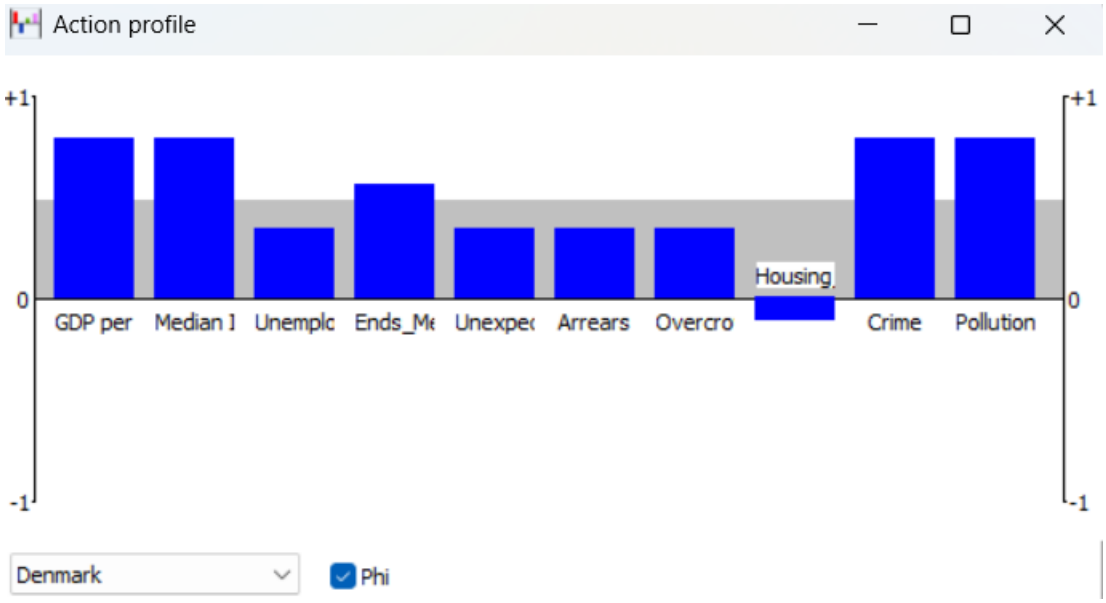


Figure 18: Denmark Action Profile, Average Scenario, Usual Function

Greece Performance Over the Years:

Table 16: Greece 2014-2023, Usual Function

Country	Year	Global_NetFlow	Rank
EL	2014	-0.662222222	99
EL	2015	-0.658484848	98
EL	2016	-0.636565657	96
EL	2017	-0.621515152	94
EL	2018	-0.572121212	92
EL	2019	-0.568989899	91
EL	2020	-0.561010101	90
EL	2021	-0.526363636	88
EL	2022	-0.46979798	85
EL	2023	-0.472626263	86

Its visualization:



Figure 19: Greece 2014-2023, Usual Function

Its indicators profile for the average scenario using Usual preference function:



Figure 20: Greece Action Profile, Average Scenario, Usual Function

Spain Performance Over the Years:

Table 17: Spain 2014-2023, Usual Function

Country	Year	Global_NetFlow	Rank
ES	2014	-0.335151515	81
ES	2015	-0.226565657	69
ES	2016	-0.227171717	70
ES	2017	-0.031515152	53
ES	2018	-0.128585859	59
ES	2019	-0.02	52
ES	2020	-0.263434343	73
ES	2021	-0.153434343	64
ES	2022	-0.137878788	61
ES	2023	-0.216565657	68

Its visualization:



Figure 21: Spain 2014-2023, Usual Function

Its indicators profile for the average scenario using Usual preference function:



Figure 22: Spain Action Profile, Average Scenario, Usual Function

Hungary Performance Over the Years:

Table 18: Hungary 2014-2023, Usual Function

Country	Year	Global_NetFlow	Rank
HU	2014	-0.757575758	100
HU	2015	-0.630909091	95
HU	2016	-0.466363636	84
HU	2017	-0.266565657	74
HU	2018	-0.129090909	60
HU	2019	-0.051313131	54
HU	2020	-0.147171717	62
HU	2021	-0.152626263	63
HU	2022	-0.072020202	55
HU	2023	0.054141414	45

Its visualization:



Figure 23: Hungary 2014-2023, Usual Function

Its indicators profile for the average scenario using Usual preference function:



Figure 24: Hungary Action Profile, Average Scenario, Usual Function

Luxembourg Performance Over the Years:

Table 19: Luxembourg 2014-2023, Usual Function

Country	Year	Global_NetFlow	Rank
LU	2014	0.341717172	31
LU	2015	0.332121212	33
LU	2016	0.27020202	36
LU	2017	0.426060606	16
LU	2018	0.40030303	19
LU	2019	0.497373737	8
LU	2020	0.381313131	23
LU	2021	0.398282828	20
LU	2022	0.537070707	5
LU	2023	0.490707071	9

Its visualization:



Figure 25: Luxembourg 2014-2023, Usual Function

Its indicators profile for the average scenario using Usual preference function:



Figure 26: Luxembourg Action Profile, Average Scenario, Usual Function

Netherlands Performance Over the Years:

Table 20: Netherlands 2014-2023, Usual Function

Country	Year	Global_NetFlow	Rank
NL	2014	0.052020202	46
NL	2015	0.160606061	39
NL	2016	0.258888889	37
NL	2017	0.428888889	15
NL	2018	0.394242424	21
NL	2019	0.455959596	12
NL	2020	0.513535354	7
NL	2021	0.529393939	6
NL	2022	0.573636364	1
NL	2023	0.557777778	3

Its visualization:

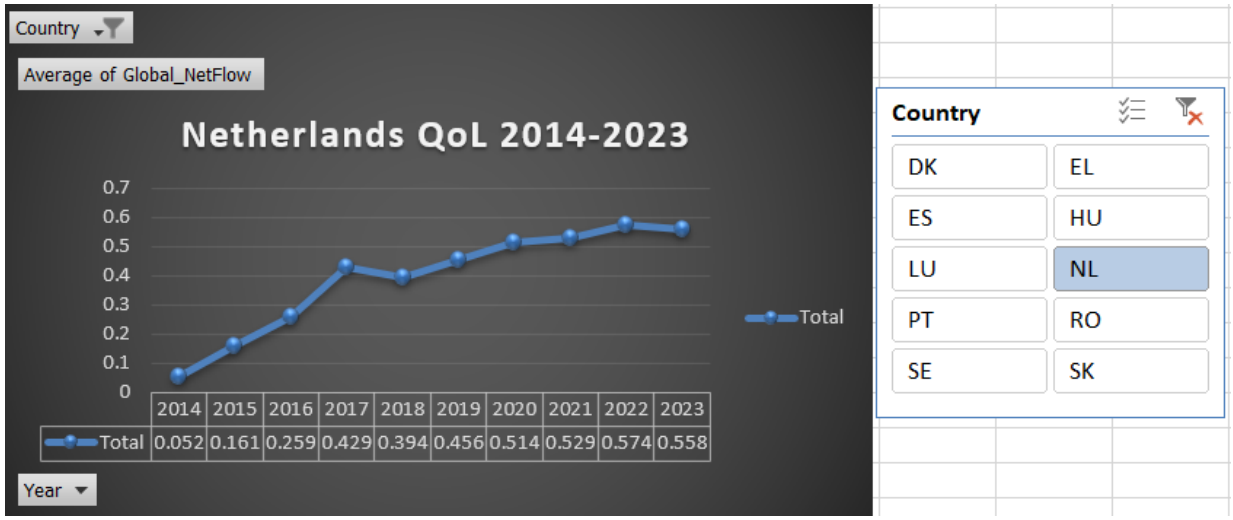


Figure 27: Netherlands 2014-2023, Usual Function

Its indicators profile for the average scenario using Usual preference function:

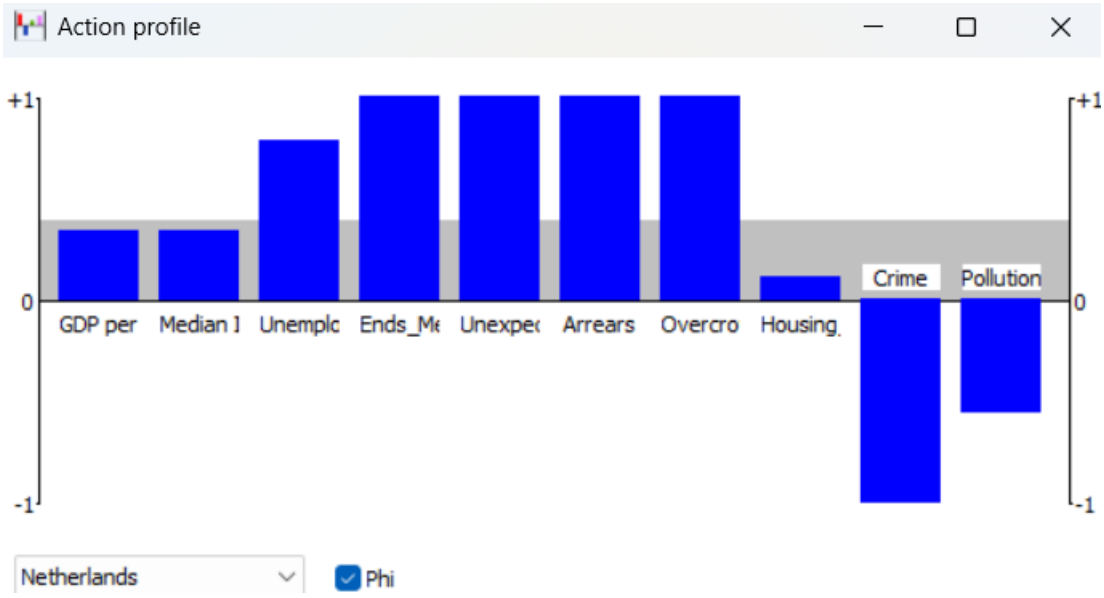


Figure 28: Netherlands Action Profile, Average Scenario, Usual Function

Portugal Performance Over the Years:

Table 21: Portugal 2014-2023, Usual Function

Country	Year	Global_NetFlow	Rank
PT	2014	-0.49030303	87
PT	2015	-0.415353535	83
PT	2016	-0.314545455	78
PT	2017	-0.243232323	71
PT	2018	-0.118383838	58
PT	2019	0.006767677	50
PT	2020	0.023636364	47
PT	2021	-0.086868687	57
PT	2022	-0.006969697	51
PT	2023	0.021515152	48

Its visualization:

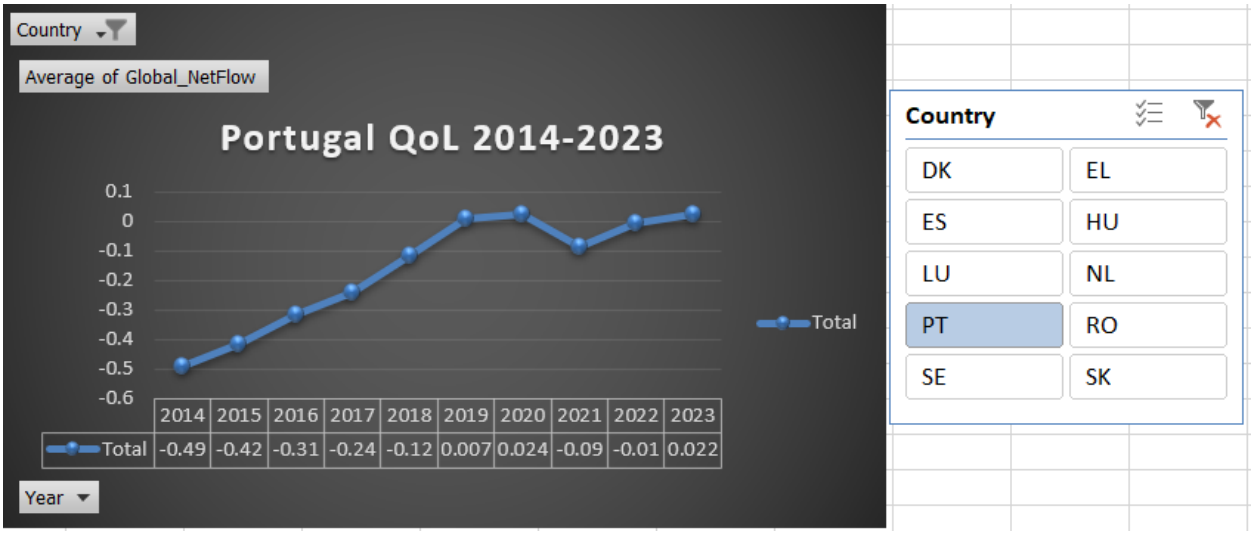


Figure 29: Portugal 2014-2023, Usual Function

Its indicators profile for the average scenario using Usual preference function:



Figure 30: Portugal Action Profile, Average Scenario, Usual Function

Romania Performance Over the Years:

Table 22: Romania 2014-2023, Usual Function

Country	Year	Global_NetFlow	Rank
RO	2014	-0.653131313	97
RO	2015	-0.596565657	93
RO	2016	-0.556161616	89
RO	2017	-0.38969697	82
RO	2018	-0.334343434	80
RO	2019	-0.252020202	72
RO	2020	-0.296565657	76
RO	2021	-0.28030303	75
RO	2022	-0.315757576	79
RO	2023	-0.164141414	66

Its visualization:

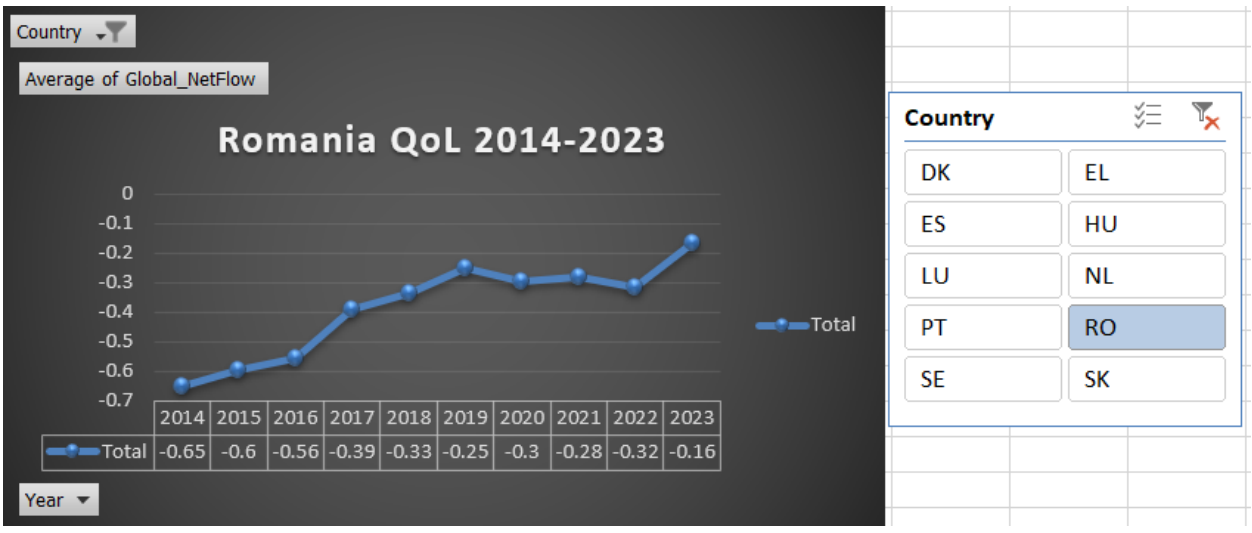


Figure 31: Romania 2014-2023, Usual Function

Its indicators profile for the average scenario using Usual preference function:



Figure 32: Romania Action Profile, Average Scenario, Usual Function

Sweden Performance Over the Years:

Table 23: Sweden 2014-2023, Usual Function

Country	Year	Global_NetFlow	Rank
SE	2014	0.3	34
SE	2015	0.355959596	27
SE	2016	0.348484848	29
SE	2017	0.432424242	14
SE	2018	0.391414141	22
SE	2019	0.36020202	26
SE	2020	0.293434343	35
SE	2021	0.360707071	25
SE	2022	0.34040404	32
SE	2023	0.342626263	30

Its visualization:



Figure 33: Sweden 2014-2023, Usual Function

Its indicators profile for the average scenario using Usual preference function:



Figure 34: Sweden Action Profile, Average Scenario, Usual Function

Slovakia Performance Over the Years:

Table 24: Slovakia 2014-2023, Usual Function

Country	Year	Global_NetFlow	Rank
SK	2014	-0.301616162	77
SK	2015	-0.201818182	67
SK	2016	-0.158181818	65
SK	2017	-0.072727273	56
SK	2018	0.02010101	49
SK	2019	0.062525253	44
SK	2020	0.129292929	40
SK	2021	0.092020202	43
SK	2022	0.097979798	42
SK	2023	0.125858586	41

Its visualization:

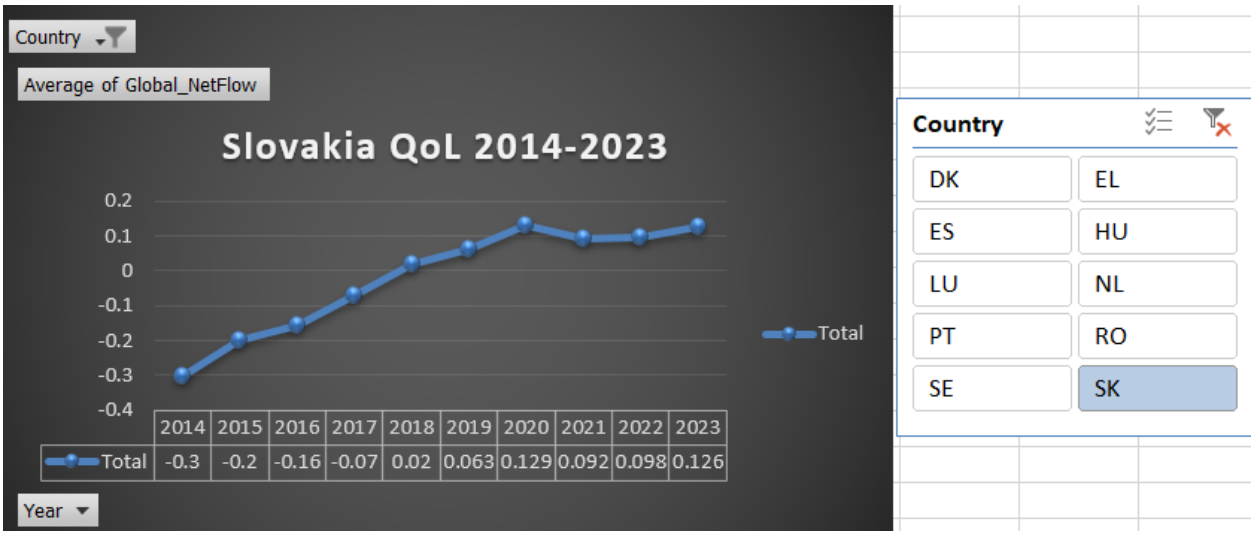


Figure 35: Slovakia 2014-2023, Usual Function

Its indicators profile for the average scenario using Usual preference function:



Figure 36: Slovakia Action Profile, Average Scenario, Usual Function

4.3 More Visual PROMETHEE

By removing the GDP per capita indicator in Visual PROMETHEE we get:

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Netherlands	0,5165	0,7582	0,2418
2	Denmark	0,4481	0,7241	0,2759
3	Sweden	0,4115	0,7057	0,2943
4	Luxemburg	0,3089	0,6545	0,3455
5	Slovakia	0,0330	0,5165	0,4835
6	Hungary	-0,1844	0,4078	0,5922
7	Spain	-0,2112	0,3944	0,6056
8	Portugal	-0,2405	0,3797	0,6203
9	Romania	-0,3748	0,3126	0,6874
10	Greece	-0,7070	0,1465	0,8535

Figure 37: PROMETHEE Flow Table without GDP per Capita

By removing the Median Income indicator in Visual PROMETHEE we get:

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Netherlands	0,5294	0,7647	0,2353
2	Denmark	0,4248	0,7124	0,2876
3	Sweden	0,4013	0,7007	0,2993
4	Luxemburg	0,2601	0,6301	0,3699
5	Slovakia	0,0745	0,5373	0,4627
6	Hungary	-0,1425	0,4288	0,5712
7	Spain	-0,2340	0,3830	0,6170
8	Portugal	-0,2497	0,3752	0,6248
9	Romania	-0,3307	0,3346	0,6654
10	Greece	-0,7333	0,1333	0,8667

Figure 38: PROMETHEE Flow Table without Median Income

By removing the Unemployment indicator in Visual PROMETHEE we get:

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Sweden	0,5190	0,7595	0,2405
2	Denmark	0,5033	0,7516	0,2484
3	Netherlands	0,4510	0,7255	0,2745
4	Luxemburg	0,4170	0,7085	0,2915
5	Slovakia	0,0353	0,5176	0,4824
6	Spain	-0,0771	0,4614	0,5386
7	Portugal	-0,1712	0,4144	0,5856
8	Hungary	-0,4562	0,2719	0,7281
9	Romania	-0,6052	0,1974	0,8026
10	Greece	-0,6157	0,1922	0,8078

Figure 39: PROMETHEE Flow Table without Unemployment

By removing the Ends Meet indicator in Visual PROMETHEE we get:

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Denmark	0,4682	0,7341	0,2659
2	Netherlands	0,4382	0,7191	0,2809
3	Sweden	0,3808	0,6904	0,3096
4	Luxemburg	0,3758	0,6879	0,3121
5	Slovakia	-0,0362	0,4819	0,5181
6	Portugal	-0,1885	0,4057	0,5943
7	Spain	-0,1910	0,4045	0,5955
8	Hungary	-0,2260	0,3870	0,6130
9	Romania	-0,3883	0,3059	0,6941
10	Greece	-0,6330	0,1835	0,8165

Figure 40: PROMETHEE Flow Table without Ends Meet

By removing the Unexpected Expenses indicator in Visual PROMETHEE we get:

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Denmark	0,4975	0,7487	0,2513
2	Netherlands	0,4318	0,7159	0,2841
3	Sweden	0,3763	0,6881	0,3119
4	Luxemburg	0,3460	0,6730	0,3270
5	Slovakia	-0,0379	0,4811	0,5189
6	Spain	-0,1616	0,4192	0,5808
7	Hungary	-0,1944	0,4028	0,5972
8	Portugal	-0,2449	0,3775	0,6225
9	Romania	-0,3838	0,3081	0,6919
10	Greece	-0,6288	0,1856	0,8144

Figure 41: PROMETHEE Flow Table without Unexpected Expenses

By removing the Arrears indicator in Visual PROMETHEE we get:

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Denmark	0,4956	0,7478	0,2522
2	Netherlands	0,4382	0,7191	0,2809
3	Sweden	0,3808	0,6904	0,3096
4	Luxemburg	0,3483	0,6742	0,3258
5	Slovakia	-0,0087	0,4956	0,5044
6	Spain	-0,1635	0,4182	0,5818
7	Hungary	-0,1985	0,4007	0,5993
8	Portugal	-0,2709	0,3645	0,6355
9	Romania	-0,3883	0,3059	0,6941
10	Greece	-0,6330	0,1835	0,8165

Figure 42: PROMETHEE Flow Table without Arrears

By removing the Overcrowding indicator in Visual PROMETHEE we get:

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Denmark	0,4870	0,7435	0,2565
2	Netherlands	0,4681	0,7340	0,2660
3	Sweden	0,4586	0,7293	0,2707
4	Luxemburg	0,3593	0,6797	0,3203
5	Slovakia	0,0284	0,5142	0,4858
6	Hungary	-0,2317	0,3842	0,6158
7	Spain	-0,2435	0,3783	0,6217
8	Portugal	-0,2506	0,3747	0,6253
9	Romania	-0,3948	0,3026	0,6974
10	Greece	-0,6809	0,1596	0,8404

Figure 43: PROMETHEE Flow Table without Overcrowding

By removing the Housing Deprivation indicator in Visual PROMETHEE we get:

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Netherlands	0,5385	0,7692	0,2308
2	Denmark	0,5360	0,7680	0,2320
3	Luxemburg	0,4408	0,7204	0,2796
4	Sweden	0,3895	0,6947	0,3053
5	Slovakia	-0,1209	0,4396	0,5604
6	Spain	-0,1453	0,4274	0,5726
7	Portugal	-0,1526	0,4237	0,5763
8	Hungary	-0,1844	0,4078	0,5922
9	Romania	-0,5287	0,2357	0,7643
10	Greece	-0,7729	0,1136	0,8864

Figure 44: PROMETHEE Flow Table without Housing Deprivation

By removing the Crime indicator in Visual PROMETHEE we get:

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Netherlands	0,6484	0,8242	0,1758
2	Sweden	0,5214	0,7607	0,2393
3	Denmark	0,4481	0,7241	0,2759
4	Luxemburg	0,4408	0,7204	0,2796
5	Slovakia	-0,1209	0,4396	0,5604
6	Spain	-0,1893	0,4054	0,5946
7	Portugal	-0,2845	0,3578	0,6422
8	Hungary	-0,3162	0,3419	0,6581
9	Romania	-0,4847	0,2576	0,7424
10	Greece	-0,6630	0,1685	0,8315

Figure 45: PROMETHEE Flow Table without Crime

By removing the Pollution indicator in Visual PROMETHEE we get:

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Netherlands	0,5326	0,7663	0,2337
2	Denmark	0,4685	0,7342	0,2658
3	Luxemburg	0,4066	0,7033	0,2967
3	Sweden	0,4066	0,7033	0,2967
5	Slovakia	-0,0378	0,4811	0,5189
6	Spain	-0,1982	0,4009	0,5991
7	Portugal	-0,2325	0,3837	0,6163
8	Hungary	-0,2486	0,3757	0,6243
9	Romania	-0,4341	0,2829	0,7171
10	Greece	-0,6632	0,1684	0,8316

Figure 46: PROMETHEE Flow Table without Pollution

Using Visual PROMETHEE the gaia plane can be observed which is crucial in analysing some of the indicators:

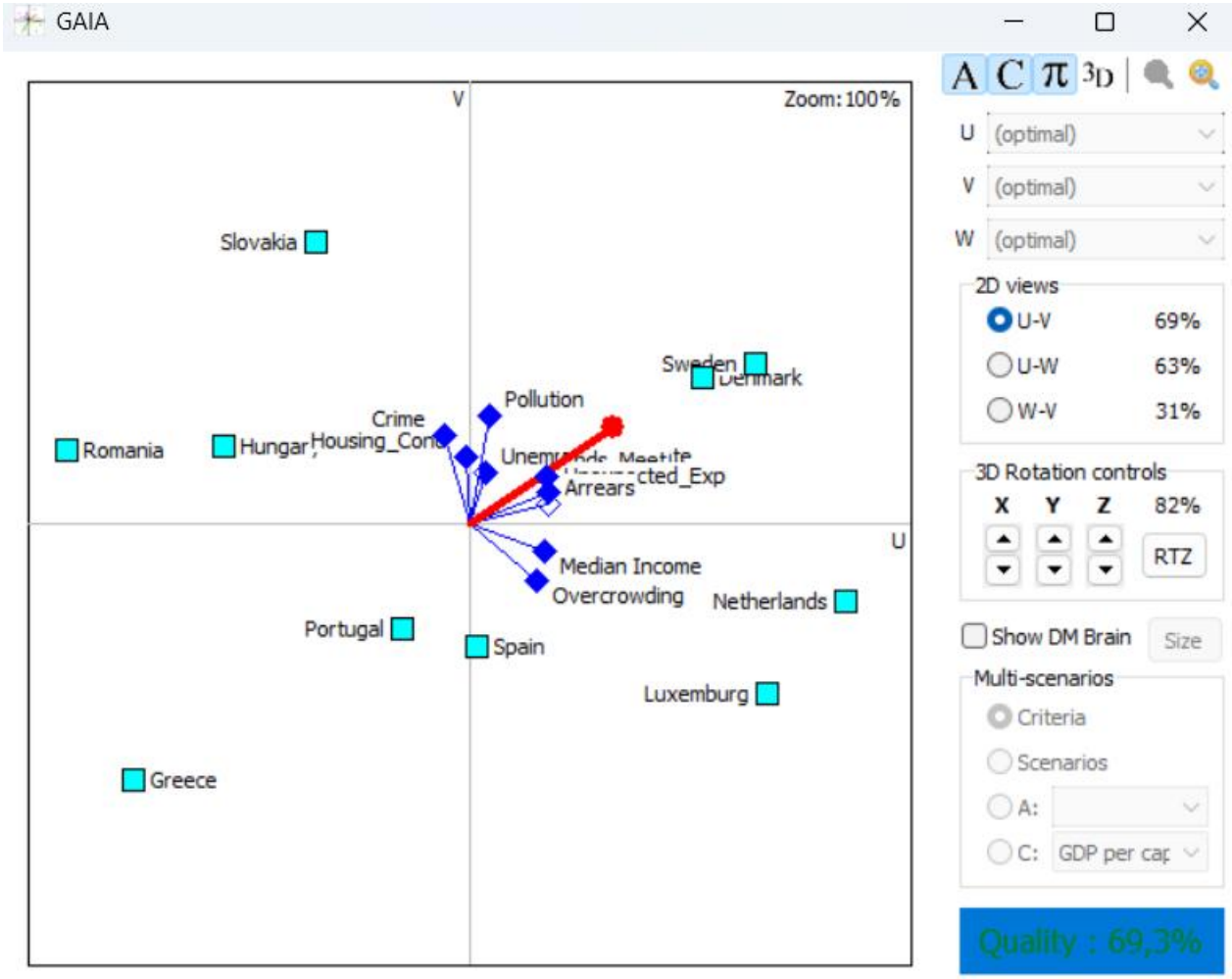


Figure 47: GAIA Plane

All countries together in a graph for the yearly scenario could be seen as follows:

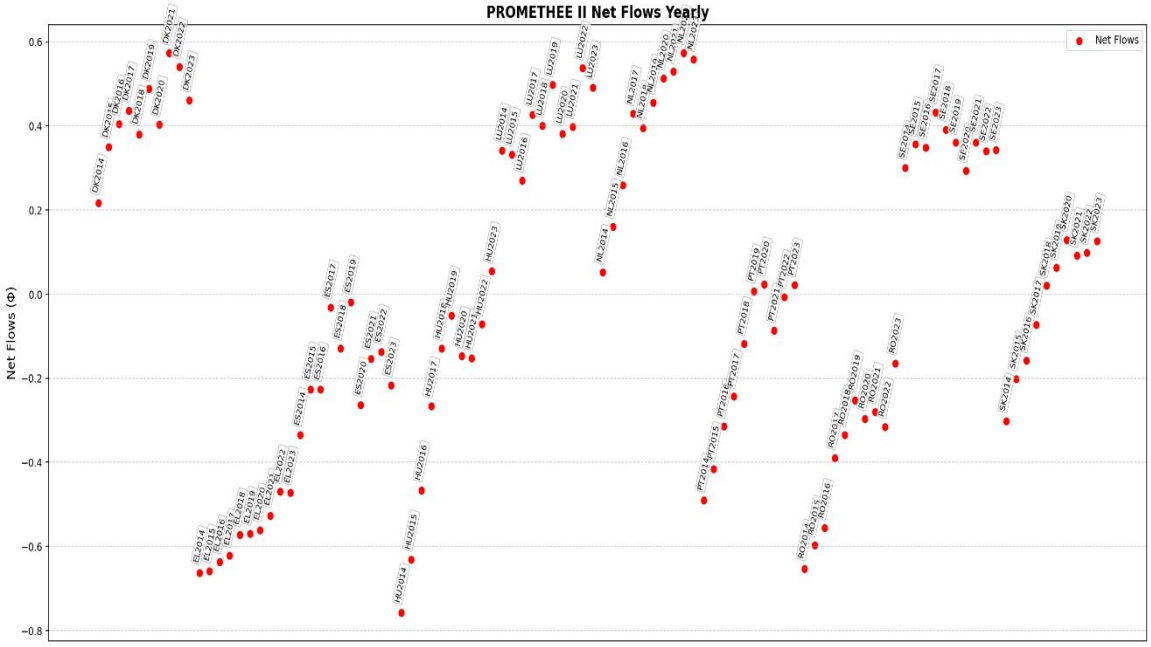


Figure 48: PROMETHEE 2014-2023 Net Flows

5. Discussion of the results

1. According to Table 2, using the **Usual Preference Function**, the country with the highest global net flow is Netherlands in 2022 with a net flow of 0.5736, while extremely close to it is Denmark in 2021 with a net flow of 0.5730. On the other hand, the country with the lowest global net flow is Hungary in 2014, with a net flow of -0.7576. This maked Hungary of 2014 the worst performing country across all years. Hungary of 2014 might be in last position but Greece has many entries in the last 10 of the ranking showing that across the years Greece might have worse Quality of Life.

   According to Table 6, using the **U-shape Preference Function,** the country with the highest global net flow is Denmark in 2022 with a net flow of 0.5250 while close to it at 2nd place is Denmark in 2021 with a net flow of 0.5185. On the other hand, the country with the lowest global net flow is again Hungary in 2014 with a net flow of -0.6931.

   According to Table 8, using the **V-shape Preference Function**, the country with the highest global net flow is Denmark in 2022 with a net flow of 0.4951, while second is Denmark of 2021 with a net flow of 0.4902. On the other hand, the country with the lowest global net flow is Greece of 2014, with a net flow of -0.6650. Hungary of 2014 still remains extremely low ranked but now it is not at last place but instead one before the last. Using the V-shape Function, Greece appears at 8/10 places in the last 10 rankings.

   According to Table 10, using the **Level Preference Function**, the country with the highest global net flow is again Denmark in 2022 with a net flow of 0.4634 while Luxembourg in 2023 is in 2nd place with close net flow value of 0.4633. On the other hand, the country with the lowest global net flow is Greece of 2014, with a net flow of -0.6610. Using this function, we see that Greece now has 9/10 appearances in the last 10 rankings. Only Greece of 2022 is missing.

   According to Table 12, using the **V-shape with indifference (Linear) Preference Function**, the country with the highest global net flow is Luxembourg in 2023 with a net flow of 0.4815, while 2nd and really close is Denmark of 2022 with net flow of 0.4711. Greece in 2014 takes the last place this time as well and again has 9/10 appearances in the last 10 rankings with only Greece of 2022 missing.

According to Table 14, using the **Gaussian Preference Function**, the country with the highest global net flow is Denmark in 2022 with a net flow of 0.4951 while 2nd is Denmark in 2021 with a net flow of 0.4902. Greece in 2014 is in the last place with a net flow of -0.6650.

2. According to Table 1, using the **Usual Preference Function**, the country with the highest ranking is the Netherlands with a global net flow of 0.5. Close to this performance is Denmark that has 0.48 global net flow. The country with the lowest performance is Greece with -0.6735 global net flow while we see Hungary having -0.24 that shows that Hungary over the years has had improvement in Quality of Life since 2014. In 2014 it had the lowest performance with a net flow of -0.7576 and in average in the decade it did -0.24.
   According to Table 5, using the **U-shape Preference function**, the country with the highest ranking is Denmark with a global net flow of 0.5033, while 2nd is Luxembourg with a net flow of 0.4822. The country with the lowest performance is Greece with -0.66 global net flow.
   According to Table 7, using the **V-shape Preference Function**, the country with the highest ranking is Denmark with a global net flow of 0.4871, while 2nd is Luxembourg with a net flow of 0.4535. The country with the lowest performance is Greece with -0.6812 global net flow.
   According to Table 9, using the **Level Preference Function**, the country with the highest ranking is Denmark with 0.4717 global net flow, while 2nd is Luxembourg with 0.4406 global net flow. The country with the lowest performance is Greece with -0.6783 global net flow.
   According to Table 11, using the **V-shape with indifference (Linear) Function**, the country with the highest ranking is Denmark with a global net flow of 0.4913, while 2nd is Luxembourg with 0.4630 global net flow. The country with the lowest performance is Greece with -0.6936 global net flow.
   According to Table 13, using the **Gaussian Preference Function**, the country with the highest ranking is Denmark with a global net flow of 0.4871, while 2nd is Luxembourg with 0.4535 global net flow. The country with the lowest performance is Greece with -0.6812 global net flow.

It appears that according to all functions except the usual, Denmark ranks first and Luxembourg 2nd. This changes when using the Usual Preference Function because by using this function there might be small differences across criteria that with using thresholds with the other functions are not considered important while in the Usual Function they are considered since even the slightest of difference matters for the preference matrix.

3. As commented in the previous question about the average scenario, changes in Preference Function can affect the rankings. This happens because the Usual Function takes into account even the smallest of differences across the criteria while the other functions have thresholds above or below which a difference is statistically significant. The changes occur mostly in the yearly scenario since there are more data that have good performance and changes can improve/worsen their performance. The only performance that never changes is the last place which is always Greece. Greece performs poorly in Quality of Life and it is easily observed by seeing the yearly scenario that in all cases of Preference Functions, Greece has many appearances in the last 10 places. The function that gives the highest 1st place ranking for the average scenario is the U-shape Preference Function with Denmark having 0.5033 global net flow, while for the yearly scenario the Usual Preference Function ranks Luxembourg 1st with 0.5736 global net flow. For the last place, the worst performance for the average scenario is when using the V-shape with indifference (Linear) Function, that has as a result Greece being last with -0.6936 global net flow. In the yearly scenario the lowest ranking is when using the Usual Preference Function that has Hungary of 2014 being in last place with -0.7576 global net flow.

4. Applying the weights that came from the questionnaire does result in different rankings. For example, according to Table 3, in the average scenario using the weights the researcher chose, the Netherlands rank 1st with a net flow of 0.5, then Denmark is second and Sweden is third. However, by using the weights from the questionnaire, Denmark becomes first with a net flow of 0.4836, Sweden goes to

second place and the Netherlands drop to the third. These changes show that participants might have given indicators, that benefit Denmark and Sweden more than the Netherlands, with higher weight values. These changes though didn't affect the lowest ranking country Greece since it is last in both cases.
In the yearly scenario with the researcher weights, according to Table 4, Netherlands in the year of 2022 is ranked first while when using the questionnaires weights Denmark of 2021 is in the first place. The last place is still for Hungary of 2014 but one thing that changes is that above Hungary now is Romania instead of Greece.

5. According to Table 1, in the average scenario, using the Usual Preference Function, the Netherlands ranks first and Greece ranks last. According to Table 2, in the yearly scenario, the best year performance is also done by the Netherlands for the year 2022. However, the worst performing country in a year is Hungary of 2014. This shows that while the highest performance country is consistent, in the lowest performance there are differences from the average scenario. This might reflect a really difficult year for Hungary in 2014 that had nothing to do with other years in the decade since in the average ranking it doesn't rank last.

6. According to the Tables 15-24:

Table 25: Worst-Best Years

Country	Worst Year	Best Year
DK	2014	2021
EL	2014	2022
ES	2014	2019
HU	2014	2023
LU	2016	2022

NL	2014	2022
PT	2014	2020
RO	2014	2023
SE	2020	2017
SK	2014	2020

A pattern emerges that shows 2014 was the worst year for Quality of Life for most countries. Only Sweden and Luxembourg are excluded from this pattern. This might reflect the aftermath of the European Union debt crisis. After 2019 many countries peaked their Quality of Life performance. This might be due to the COVID-19 era and the effort to improve economic stability again after this global pandemic.

7. According to the Tables 15-24, for countries like Denmark, Spain, Hungary, Luxembourg, Sweden and Romania in 2020 Quality of Life was reduced and in 2021 the countries performed better again.
   The COVID-19 period had a huge impact on Quality of Life across many EU countries. This is also reflected in this PROMETHEE II research by seeing the net flows of 2019,2020 and 2021. This shows that in 2020 there was problem in indicators of economic stability, employment and more. This drop reflects the consequences of lockdowns, restrictions and uncertainty on the economic part of Quality of Life. Some countries though were not affected such as Greece, Netherlands, Portugal and Slovakia. Some improved their Quality of Life score. Romania for example peaked its performance in 2020.

8. According to Tables 15-24, the country that showed the most significant improvement over the decade is Hungary. In 2014 Hungary had a net flow of -0.7576 ranking last among all other countries. By 2023, Hungary improved its well-being by having a net flow of 0.054. This +0.8116 change shows a consistent improvement across the years and indicated an improvement in all the indicators used.

The least improvement was made from Sweden since in 2014 it had a net flow of 0.3 and in 2023 Sweden had 0.343. Sweden has a great score for Quality of Life over the decade and this shows stability even though big improvement wasn't done.

9. While GDP per capita as said before, has been viewed as a crucial determinant of well-being, our results according to Table 1, show that countries with similar GDP levels do not always achieve similar Quality of Life rankings. Denmark (GDP 32.843) and Sweden (GDP 30.795) have close GDP values and perform similarly in our rankings since they have net flows of 0.478 and 0.425. In these 2 high-income countries correlation exists between GDP and QoL rankings. However, when reviewing countries of lower economy, the correlation weakens. Portugal (GDP 13788) and Greece (GDP 13379) have GDP values nearly the same, yet their ranking is completely different. Portugal has a net flow of -0.230 and Greece -0.67 being the last country in ranking. Similarly, Hungary (GDP 7769) and Romania (GDP 6885) are close in their GDP per capita indicator but Hungary ranks higher with a net flow of -0.238 while Romania has net flow of -0.43. Our research shows that Quality of Life is a complex concept that many indicators can affect and even though some countries might have similar GDP profiles, they do not always rank similarly.

10. According to Table 1 and Figure 62, using the Usual Preference Function, the results obtained from the Python script are fully consistent with the ones generated by Visual PROMETHEE. As seen in the outputs, the global net flows and final rankings for all countries are the same in both cases. This confirms that our Python script was correct, validating that the PROMETHEE II algorithm was accurately performed.

11. The results of this study show that some countries might have high performance in some indicators but extremely low at others. This shows how multidimensional Quality of Life is and how weighting the criteria differently can affect the final ranking. For example:

    a) According to Figure 24, Hungary performs poorly in indicators such as GDP per capita, Median Income and Housing Deprivation. However, in the Unemployment criteria, it performs the best across all other countries.

    b) According to Figure 26, Luxembourg, is the best in GDP per capita and Median Income criteria across all other countries but performs badly in the Housing Deprivation and Crime criteria and extremely bad at the Pollution indicator.

    c) According to Figure 28, Netherlands, is the best in Ends Meet, Unexpected Expenses, Arrears and Overcrowding indicators but performs poorly in the Crime criteria since it performs the worst in this compared to all other countries.

    This shows why single indicator analyses can be misleading since some countries may perform poorly in some indicators but might excel in others and vise versa.

12. To determine the most crucial indicators, we observe the GAIA plane, a graphical representation of the PROMETHEE analysis. The closer the blue vectors are to the red decision axis, the greater its impact on the final ranking. According to Figure 47, the indicators with the most impact are:

    a) Arrears

    b) Unexpected Expenses

    c) Unemployment

    Indicators like Crime, Housing Deprivation, Pollution are positioned far from the decision axis and have less influence in the final rankings.

13. According to Figure 47, 2 of our criteria appear to be strongly correlated. This shows that they behave in the same way in how they influence the final rankings. By observing the GAIA plane, only Median Income is visible and not GDP per capita. By removing from the indicators, the Median Income and only letting GDP per capita, we see that GDP per capita is in the exact same position as the Median Income vector used to be. This suggests a high correlation between these 2 indicators and is the reason why Visual PROMETHEE visualizes in the GAIA plane only one of them.

14. By removing one criterion at a time, we observe that many rankings can change especially in middle categories.

a) According to Figure 37, when GDP is removed, instead of having Spain, Portugal and Hungary at the 5,6,7 ranks we see a new order of Hungary, Spain and Portugal

b) According to Figure 38, when Median Income is removed, the same thing happens with Hungary, Spain and Portugal being at places 5,6,7. This also shows how GDP and Median Income indicators correlate as we said before using the GAIA plane.

c) According to Figure 39, when Unemployment is removed, there is change in the first 3 places. Instead of having Netherlands, Denmark and Sweden at places 1,2,3 we now have Sweden, Denmark and Netherlands.

d) According to Figure 40, when removing the Ends Meet indicator, Netherlands and Denmark from places 1,2 switch places and now Denmark is first in the ranking while Netherlands is second. Spain and Portugal also switch places from 6,7 and now Portugal is sixth while Spain is seventh.

e) According to Figure 41, when removing the Unexpected Expenses indicator, the same thing happens with removing the Ends Meet criteria, and now Denmark, Netherlands, Portugal and Spain are ranked 1,2,6,7

f) According to Figure 42, when removing Arrears, Denmark switches to first place while Netherlands to second and Portugal goes to 8th place while hungary to 7th.

g) According to Figure 43, when removing Overcrowding, Denmark switches to first place while Netherlands to second and Hungary, Spain and Portugal are at places 6,7,8.

h) According to Figure 44, when removing Housing Deprivation, the change that occurs is that at places 3,4 we used to have Sweden, Luxembourg but now the opposite happens since we have Luxembourg at 3rd place and Sweden at 4th.

i) According to Figure 45, when removing Crime, Denmark and Sweden switch places and now we have Sweden as 2nd and Denmark as 3rd.

j) According to Figure 46, when removing Pollution, something different happens. We used to have Sweden at 3rd place and Luxembourg at 4th but now

both of them rank at 3rd place with them performing exactly the same and having the exact same global netflows.

The only countries that remain in the exact same positions regardless of which criterion was removed, is Romania (9th place) and Greece (10th place). This shows that they perform poorly across all indicators.

15. According to Figure 10, using the Usual Preference Function for the Average scenario the countries that are ranked first and last in each indicator are:

Table 26: Worst-Best Countries Across Indicators, Average Scenario, Usual Function

	Best	**Worst**
GDP per Capita	Luxembourg	Romania
Median Income	Luxembourg	Romania
Unemployment	Hungary	Greece
Ends Meet	Netherlands	Greece
Unexpected Expenses	Netherlands	Greece
Arrears	Netherlands	Greece
Overcrowding	Netherlands	Romania
Housing Deprivation	Slovakia	Portugal
Crime	Slovakia	Netherlands
Pollution	Sweden	Greece

This table helps us observe how no single country dominates across all criteria. There are 5 (half of the selected) countries that excel in the indicators. This shows how complex Quality of Life is as a concept and how one country can excel at some factors but in others not so much. Also, it doesn't matter if one country only excels in some of the factors but the overall performance. This is easily observed

also in the case of Slovakia and Luxembourg that both excel in 2 indicators each but their difference in the final ranking is huge with one being close to 0 and the other close to 0.4

6. Conclusion

This research was done to observe and compare Quality of Life across several European Union countries over a 10-year period (2014-2023). A Multi-Criteria Decision Analysis method (PROMETHEE II) was performed. For the analysis, 2 scenarios were compiled. One for aggregated data using the mean values of every criterion over the 10-year period so that we could compare the overall average performance of each country and the other for a yearly comparison using every country in every year of this decade. While comparing the countries performance, six difference preference functions were used, alongside both researcher-assigned weights and weights that came from a questionnaire that 100 university students and graduates answered.

The findings demonstrated that countries such as Denmark, the Netherlands and Sweden are consistent in maintaining the first 3 places in the final ranking while Greece and Romania rank most of the times the lowest. A tough year for most of the countries was 2014 since most of them had their lowest well-being scores in this year and a crucial observation was that Hungary, despite starting from the lowest ranking in 2014, had the most improvement in the decade. By using all 6 preference functions, we saw that rankings are affected especially in middle-ranking countries. There were cases though that many of the preference functions had similar outputs and only contradicted with the usual function. This happened because the usual function takes into account even the smallest difference between 2 criteria when computing the preference between 2 countries while this doesn't happen in the other functions because thresholds exist above/bellow which a difference is statistically significant. The weights used came from the researchers view and from the answers of a questionnaire that was designed to assess the significancy of each indicator on the participants perspective. There were differences on the outputs and this happened because by evaluating each indicator differently, the final rankings are affected since we proved that there are countries that excel in some indicators but perform poorly in others. This also showed why it is important to use a multidimensional approach when evaluating Quality of Life but also being careful in how each indicator is weighted depending on what criteria we think is most crucial in well-being. A sensitivity analysis showed that when removing only one indicator changes occur in middle and upper positions of the final ranking. However, Romania and Greece consistently remain at the bottom of the final rankings having the places 9th and 10th showing poor performance across most criteria.

Overall, this research shows the value of the PROMETHEE II methodology in handling complex concepts such as Quality of Life.

7. Limitations

During the course of this research, some limitations occurred. Firstly, while both scenarios (yearly, average) were analyzed, the first one was significantly larger. This made it difficult to display a full step-by-step PROMETHEE II methodology in excel like we displayed for the average scenario. The full process, for the yearly scenario was only demonstrated in the excel for one criterion to ensure that the Python scrip was executed correctly. Secondly, regarding the weight scenarios, the initial plan was to compare 2 different scenarios. The first one was having equal weights across indicators and the second having the weights from the questionnaire. However, most participants rated many indicators with a significance close to 0.10, leading to results that were almost identical to the ones produced from the equal weights scenario. As a result, the first scenario (equal weights), was then changed with weights based on the researcher's personal judgement. Lastly, one more problem that occurred was when choosing countries and indicators or as we described them in our methodology, actions and alternatives. Quality of Life is a complex concept with many factors so deciding on which countries and especially indicators to choose was difficult. Even the 10 indicators we chose might seem enough but aren't even close to translating Quality of Life fully.

8. Recommendations

1. Expand the Survey sample for Weight Determination

   Future studies should conduct a larger-scale survey. Doing this, a greater number of participants could take place and this would allow for a more representative result in creating weights for each of the indicators. This could maybe solve the problem that occurred on this thesis that was that the weights of the questionnaire were almost similar to the equal weight scenario.

2. More expanding Set of Indicators

   This study used 10 indicators focused on economic, material, criminality and pollution aspects. It would be wise to expand the criteria to include more dimensions, such as physical health, mental well-being, access to services and more. As we mentioned in the limitations, Quality of Life is so complex that even our 10 criteria are not enough to describe it as a concept. By having more indicators and from different dimensions, this would broaden our view in Quality of Life since we saw how multi-dimensional it is.

3. Include all EU Countries

   This research used 10 different European Union countries that scored differently across the indicators chosen. It could be scaled up to cover all members of the European Union to observe the overall Quality of Life ranking in Europe.

4. Quality of Life Over Time with Event Context

   As seen in the outputs, both 2014 and 2020 were periods that the countries performed poorly. This happened because in the first case it was a year that many countries suffered from the post European Economic Crisis and in the second case because of the effects COVID-19 had in our lives. It would be wise in future researches to perform analysis based on these years to better understand how European Union Economic Crisis, COVID-19 or other contexts affected Quality of Life.

5. PROMETHEE vs Other MCDA Methods

   It would be wise to extend this research in using other MCDA methods such as TOPSIS, ELECTRE and the others we mentioned so that we could compare the results and see the differences between them.

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10. Appendices

Appendix A: Questionnaire

Quality of Life Criteria Ranking

This survey aims to understand how different factors influence quality of life in a country. You will be asked to rank 10 different criteria based on their importance in determining quality of life.

Your participation in this survey is completely voluntary. All responses will be kept strictly anonymous and will not be linked to any personal information. The data collected will be used only for academic purposes, specifically as part of a Master's thesis on quality of life in EU countries.

Instructions for Participants:

- You will be ranking **10 different criteria** based on their importance in determining **quality of life**.
- Assign a ranking number from 1 (Not Important) to 5 (Very Important) to each criterion.
- Please read each question carefully before selecting your rankings.

* Υποδεικνύει απαιτούμενη ερώτηση

1. C.1 Country's economic output per person (GDP per Capita)? *

Να επισημαίνεται μόνο μία έλλειψη.

	1	2	3	4	5	
Not	◯	◯	◯	◯	◯	Very Important

2. C.2 Median income level in a country? *

Να επισημαίνεται μόνο μία έλλειψη.

	1	2	3	4	5	
Not	◯	◯	◯	◯	◯	Very Important

Figure 49: Questionnaire part a

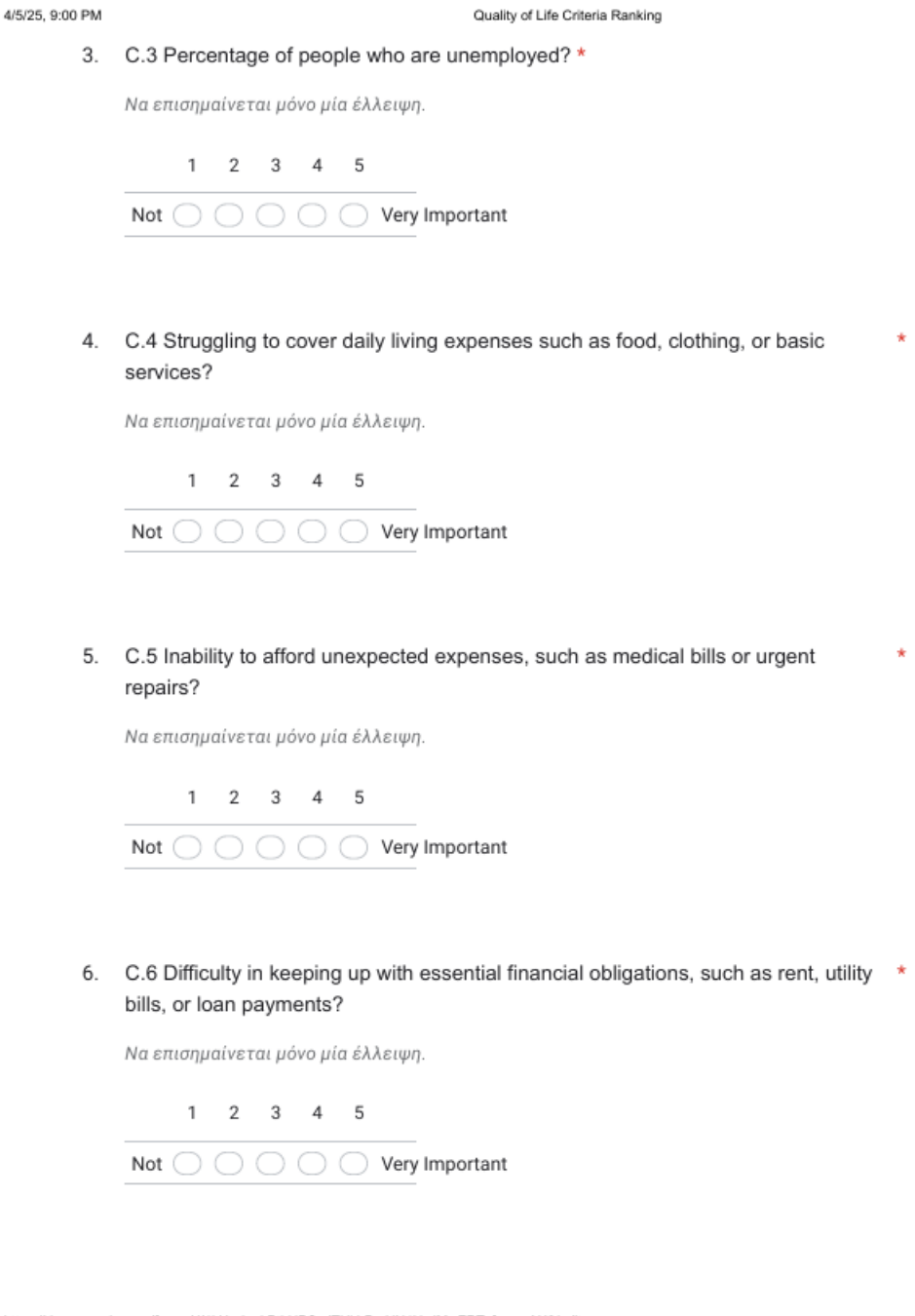


Figure 50: Questionnaire part b

7. C.7 Living in a home with too many people for the available space? *

Να επισημαίνεται μόνο μία έλλειψη.

	1	2	3	4	5	
Not	◯	◯	◯	◯	◯	Very Important

8. C.8 Living in poor housing conditions, such as lack of heating, sanitation, or structural issues? *

Να επισημαίνεται μόνο μία έλλειψη.

	1	2	3	4	5	
Not	◯	◯	◯	◯	◯	Very Important

9. C.9 Level of crime, violence, and vandalism in a country? *

Να επισημαίνεται μόνο μία έλλειψη.

	1	2	3	4	5	
Not	◯	◯	◯	◯	◯	Very Important

10. C.10 Exposure to pollution, such as poor air quality, noise, or contaminated water? *

Να επισημαίνεται μόνο μία έλλειψη.

	1	2	3	4	5	
Not	◯	◯	◯	◯	◯	Very Important

Figure 51: Questionnaire part c

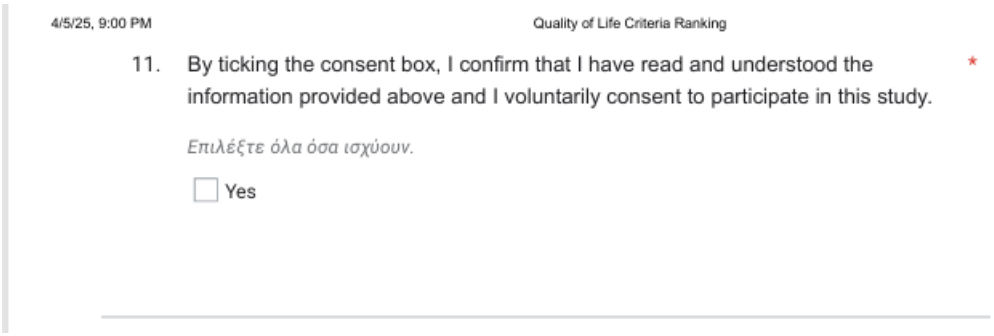


Figure 52: Questionnaire part d

Appendix B: Questionaire Results

Χρονική σήμανση	C.1	C.2	C.3	C.4	C.5	C.6	C.7	C.8	C.9	C.10
3-26-2025 19:05:58	5	5	4	3	2	4	2	2	5	1
3-26-2025 19:56:28	3	4	4	5	5	4	3	5	5	5
3-26-2025 21:40:44	4	2	5	3	2	4	2	2	3	1
3-26-2025 23:45:12	2	5	5	5	5	3	2	1	4	1
3-27-2025 6:13:27	3	3	4	4	3	3	4	4	3	3
3-27-2025 6:51:13	4	4	5	5	3	3	2	2	3	2
3-27-2025 12:45:18	5	4	5	5	5	5	4	5	4	4
3-29-2025 15:07:20	5	5	5	5	5	5	1	5	5	4
3-29-2025 15:13:48	4	4	4	5	5	5	3	5	5	4
3-29-2025 15:13:51	3	3	5	5	4	5	5	5	5	4
3-29-2025 15:23:31	5	4	4	5	4	3	4	4	5	5
3-29-2025 15:23:56	5	3	4	4	4	5	3	4	3	3
3-29-2025 15:27:27	5	5	5	5	5	5	4	4	4	4
3-29-2025 15:28:35	5	5	5	5	5	5	1	5	5	3
3-29-2025 15:30:38	5	2	5	5	4	4	2	3	5	4
3-29-2025 15:36:44	4	4	3	5	5	4	2	4	4	4
3-29-2025 15:43:22	4	4	3	5	5	5	2	5	3	4
3-29-2025 15:47:11	3	3	2	5	5	5	3	5	4	5
3-29-2025 15:47:28	3	3	5	3	5	4	3	5	5	5
3-29-2025 15:49:14	4	4	4	5	5	5	4	5	5	5
3-29-2025 15:55:27	3	5	3	5	3	5	3	4	3	5
3-29-2025 16:04:21	5	5	5	5	4	5	3	3	4	4
3-29-2025 16:04:41	3	4	5	5	4	4	3	4	3	3
3-29-2025 16:06:53	4	4	3	4	5	5	2	3	3	5
3-29-2025 16:10:23	5	5	4	4	4	5	4	5	4	5
3-29-2025 16:12:05	4	4	4	5	5	5	4	5	4	4
3-29-2025 16:19:31	5	5	5	5	5	5	4	5	5	5
3-29-2025 16:27:46	4	5	3	5	5	5	2	5	3	2
3-29-2025 16:29:47	3	3	4	2	3	3	1	2	4	2
3-29-2025 16:32:49	4	5	4	5	5	5	5	5	4	4
3-29-2025 16:36:30	3	5	5	5	3	4	2	5	4	2
3-29-2025 16:45:40	4	5	3	4	5	4	3	4	5	4

Figure 53: Questionnaire Results part a

Χρονική σήμανση	C.1	C.2	C.3	C.4	C.5	C.6	C.7	C.8	C.9	C.10
3-29-2025 16:58:35	1	3	2	5	4	5	3	4	1	5
3-29-2025 17:15:15	4	3	5	5	2	4	3	5	5	3
3-29-2025 17:21:46	3	5	3	5	4	4	3	3	5	4
3-29-2025 17:30:41	4	4	4	5	5	4	2	3	3	3
3-29-2025 17:31:47	3	4	3	5	5	4	2	4	2	2
3-29-2025 17:43:55	3	4	2	4	3	5	2	3	5	2
3-29-2025 18:10:05	4	3	3	5	5	5	4	5	4	3
3-29-2025 18:18:17	5	5	3	5	4	4	3	1	3	3
3-29-2025 19:22:30	2	2	5	5	4	3	2	2	2	1
3-29-2025 20:10:23	5	3	3	5	4	3	3	3	2	2
3-29-2025 20:20:05	3	4	2	5	4	3	2	5	5	3
3-29-2025 20:54:05	3	1	3	5	4	5	2	3	1	2
3-29-2025 21:01:03	3	5	2	5	3	3	2	4	2	2
3-29-2025 21:01:43	1	1	1	5	5	5	1	3	2	2
3-29-2025 21:02:02	2	2	3	4	5	5	1	5	2	5
3-29-2025 21:05:26	1	5	4	5	5	5	3	5	1	1
3-29-2025 21:34:45	3	4	2	5	5	5	3	5	2	3
3-29-2025 22:21:14	3	5	5	5	4	3	3	5	5	3
3-29-2025 23:21:21	3	4	5	4	3	3	2	4	2	3
3-30-2025 0:03:53	4	3	2	4	4	5	2	4	3	2
3-30-2025 0:29:19	4	5	5	3	4	4	2	2	2	1
3-30-2025 1:26:14	2	3	4	4	3	3	3	3	4	2
3-30-2025 4:06:17	5	4	4	5	5	4	3	5	4	4
3-30-2025 4:19:20	3	3	3	4	4	5	3	4	2	3
3-30-2025 4:44:06	4	3	2	5	4	3	2	1	3	3
3-30-2025 13:50:09	4	4	3	5	3	4	2	4	3	3
3-30-2025 14:59:11	3	3	3	4	3	3	4	3	3	4
3-30-2025 15:02:41	4	4	5	5	5	4	3	4	2	4
3-30-2025 15:03:17	5	3	5	5	5	5	4	5	4	4
3-30-2025 15:42:46	4	4	5	3	1	2	3	2	3	3
3-30-2025 15:48:24	3	5	5	5	5	5	5	5	2	4
3-30-2025 15:48:31	5	5	3	5	5	5	5	5	4	5

Figure 54: Questionnaire Results part b

Χρονική σήμανση	C.1	C.2	C.3	C.4	C.5	C.6	C.7	C.8	C.9	C.10
3-30-2025 16:01:30	4	3	2	4	4	4	2	4	3	4
3-30-2025 16:03:38	4	4	5	5	5	5	2	5	4	5
3-30-2025 16:06:55	4	3	5	5	5	5	4	4	5	5
3-30-2025 16:39:09	5	4	3	5	4	4	3	4	3	2
3-30-2025 16:42:55	5	3	2	3	3	2	2	4	5	5
3-30-2025 16:43:35	3	3	2	4	4	3	4	2	4	3
3-30-2025 16:45:18	5	4	4	5	4	4	1	3	4	3
3-30-2025 16:50:48	4	3	3	5	4	5	2	3	4	2
3-30-2025 16:50:54	4	3	3	5	3	4	5	4	5	4
3-30-2025 16:51:43	4	3	5	3	5	2	1	3	3	5
3-30-2025 16:54:50	2	4	3	5	5	5	2	5	4	3
3-30-2025 16:55:23	4	3	5	2	4	2	2	3	4	3
3-30-2025 16:56:40	5	4	3	5	4	3	2	3	4	4
3-30-2025 17:05:42	4	2	4	5	3	4	2	4	4	3
3-30-2025 17:07:00	3	2	4	5	5	3	1	4	2	2
3-30-2025 17:31:27	2	2	3	3	4	5	2	5	3	4
3-30-2025 17:44:53	4	5	4	2	4	4	2	5	4	2
3-30-2025 17:53:31	3	5	2	4	5	3	3	1	4	5
3-30-2025 17:56:05	4	4	4	4	5	4	5	3	3	3
3-30-2025 17:57:29	3	3	4	5	3	5	4	4	4	4
3-30-2025 18:01:09	4	4	3	5	5	4	2	4	5	3
3-30-2025 18:28:26	3	3	5	5	4	4	3	5	4	4
3-30-2025 18:57:20	4	4	2	2	4	3	3	4	3	3
3-30-2025 19:09:11	5	4	3	5	5	4	5	5	2	2
3-30-2025 19:14:41	3	4	5	3	5	5	3	4	5	3
3-30-2025 19:25:07	4	4	5	5	5	5	5	5	4	5
3-30-2025 19:39:54	4	4	3	4	5	5	3	4	5	5
3-30-2025 19:45:09	2	2	2	5	5	5	4	5	3	5
3-30-2025 20:03:55	5	5	5	3	3	5	2	3	4	3

Figure 55: Questionnaire Results part c

Χρονική σήμανση	C.1	C.2	C.3	C.4	C.5	C.6	C.7	C.8	C.9	C.10
3-30-2025 20:20:42	1	4	5	5	5	5	1	1	2	1
3-30-2025 20:21:20	3	2	3	5	5	5	2	2	1	1
3-30-2025 20:21:54	3	4	3	5	5	3	2	2	4	2
3-30-2025 20:22:14	2	2	3	4	4	5	2	3	1	2
3-30-2025 20:39:53	2	4	5	5	5	5	5	5	4	5
3-30-2025 21:15:38	4	3	4	5	5	3	2	4	4	5
3-30-2025 21:29:57	4	4	3	5	4	4	1	3	3	2

Figure 56: Questionnaire Results part d

Appendix C: Python Code

```
import pandas as pd
import numpy as np
import Eurostat
from graphviz import Digraph
import matplotlib.pyplot as plt
import math


def process_dataset(dataset_code, filters, value_name, countries):

    data = eurostat.get_data_df(dataset_code)

    for column, condition in filters.items():
        if column in data.columns:
            print(f"Filtering {column} == {condition}")
            data = data[data[column] == condition]

    years = [str(year) for year in range(2014, 2024)]
    years_available = [col for col in years if col in data.columns]

    data = pd.melt(data, id_vars=['geo\\TIME_PERIOD'], value_vars=years_available,
var_name='Year', value_name=value_name)
   data.rename(columns={'geo\\TIME_PERIOD': 'Country'}, inplace=True)
   data['Year'] = data['Year'].astype(int)
   data = data[data['Country'].isin(countries)]

   return data
```

```
countries = ['DK', 'EL', 'ES', 'LU', 'NL', 'PT', 'RO', 'SK', 'SE', 'HU']
criteria_info = {
'C.1 Median_Income': ('ilc_di03', {'unit': 'EUR', 'age': 'TOTAL', 'sex': 'T', 'indic_il':
'MED_E'}),
'C.2 GDP_Per_Capita': ('nama_10_pc', {'unit': 'CLV15_EUR_HAB', 'na_item': 'P41'}),
'C.3 Ends_Meet': ('ilc_mdes09', {'subjnmon': 'EM_GD', 'hhtyp': 'TOTAL', 'incgrp': 'TOTAL'}),
 'C.4 H_Deprivation': ('ilc_mdho01', {'hhtyp': 'TOTAL', 'sex': 'T', 'age': 'TOTAL', 'incgrp':
'TOTAL'}),
 'C.5 H_Overcrowding': ('ilc_lvho05a', {'incgrp': 'TOTAL', 'age': 'TOTAL', 'sex': 'T'}),
 'C.6 Unemployment': ('lfsa_urgaed', {'sex': 'T', 'age': 'Y15-74', 'isced11': 'TOTAL'}),
 'C.7 Unexpected_Expenses': ('ilc_mdes04', {'hhtyp': 'TOTAL', 'incgrp': 'TOTAL'}),
 'C.8 Arrears': ('ilc_mdes05', {'hhtyp': 'TOTAL', 'incgrp': 'TOTAL'}),
 'C.9 Criminality': ('ilc_mddw03', {'hhtyp': 'TOTAL', 'incgrp': 'TOTAL'}),
'C.10 Pollution': ('ilc_mddw02', {'hhtyp': 'TOTAL', 'incgrp': 'TOTAL'})
}

datasets = []
for criterion, (dataset_code, filters) in criteria_info.items():
        print(f" Processing {criterion}...")
        df = process_dataset(dataset_code, filters, criterion, countries)
        datasets.append(df)

combined_data = datasets[0]
for dataset in datasets[1:]:
        combined_data = combined_data.merge(dataset, on=['Country', 'Year'],
how='outer')

for column in combined_data.columns[2:]:
        combined_data[column] =
combined_data.groupby('Country')[column].transform(lambda x: x.fillna(x.mean()))

aggregated_data = combined_data.groupby("Country").mean().reset_index()
criteria_columns = list(criteria_info.keys())
aggregated_data = aggregated_data[['Country'] + criteria_columns]
```

```
criteria = {
        'C.1 Median_Income' : 'max' ,
        'C.2 GDP_Per_Capita': 'max' ,
        'C.3 Ends_Meet': "min" ,
        'C.4 H_Deprivation': "min" ,
        'C.5 H_Overcrowding': "min" ,
        'C.6 Unemployment': "min" ,
        'C.7 Unexpected_Expenses': "min" ,
        'C.8 Arrears': "min" ,
        'C.9 Criminality': "min" ,
        'C.10 Pollution': "min" ,
 }

c = [1 if criteria[crit] == 'max' else 0 for crit in criteria.keys()]

valid_functions = ['u', 'us', 'vs', 'le', 'li', 'g']
print("\n Available Preference Functions:", valid_functions)

while True:
        user_input = input("Select a preference function for all criteria ") .strip().lower()
        if user_input in valid_functions:
                break
         else:
                print(" Invalid choice, please choose a correct preference function ")


f = [user_input] * len(criteria)
w = np.array([0.15, 0.09, 0.11, 0.09, 0.06, 0.15, 0.12, 0.11, 0.09, 0.03])
print("\n Weights per Criterion:")
for crit, weight in zip(criteria.keys(), w):
        print(f"{crit} has weight: {weight:.4f}")



def unical(diff_value, q, p, s, f):
```

```
        if f == 'u':
                return 1 if diff_value > 0 else 0
        elif f == 'us':
                return 1 if diff_value > q else 0
        elif f == 'vs':
                if diff_value <= 0:
                        return 0
                elif diff_value > p:
                        return 1
                else:
                        return diff_value / p
        elif f == 'le':
                if diff_value <= q:
                        return 0
                elif diff_value > q and diff_value <= p:
                        return 0.5
                else:
                        return 1
        elif f == 'li':
                if diff_value <= q:
                        return 0
                elif diff_value > p:
                        return 1
                else:
                        return (diff_value - q) / (p - q)
        elif f == 'g':
                return 1 - math.exp(-((diff_value) ** 2) / (2 * (s ** 2))) if diff_value > 0
else 0
         else:
                raise ValueError("Invalid preference function type")
```

```
def compute_differences(x, c):
        diff_matrices = {}
        for i, criterion in enumerate(criteria):
                if c[i]:
                        diff_matrix = x[:, i:i+1] - x[:, i]
                else:
                        diff_matrix = x[:, i] - x[:, i:i+1]
                diff_matrices[criterion] = diff_matrix
        return diff_matrices

def compute_thresholds(data):
        data = data[[col for col in data.columns if col in criteria]]


        q_values = data.std() * 0.25
        p_values = data.std() * 0.75
        s_values = data.std() * 0.45

        return np.array([q_values, p_values, s_values])

def compute_preference_matrix(diff_matrices, q, p,s, f):
        pref_matrices = {}
        for i, (criterion, diff_matrix) in enumerate(diff_matrices.items()):
                pref_matrix = np.zeros(diff_matrix.shape)
                for row in range(diff_matrix.shape[0]):
                        for col in range(diff_matrix.shape[1]):
                                pref_matrix[row, col] = unical(diff_matrix[row, col], q[i],
p[i],s[i], f[i])
                pref_matrices[criterion] = pref_matrix
        return pref_matrices

def compute_flows(preference_matrices):
        return {criterion: {
                                "Positive_Flow": pref_matrix.sum(axis=1) / (len(pref_matrix)
- 1),
```

```
                                "Negative_Flow": pref_matrix.sum(axis=0) /
(len(pref_matrix) - 1),
                                "Net_Flow": pref_matrix.sum(axis=1) / (len(pref_matrix) - 1)
- pref_matrix.sum(axis=0) / (len(pref_matrix) - 1)}
                for criterion, pref_matrix in preference_matrices.items()}

def compute_weighted_flows(unicriterion_flows, weights):
        weighted_flows = {f"Weighted_{criterion}": weights[i] * flows["Net_Flow"]
                                for i, (criterion, flows) in
enumerate(unicriterion_flows.items())}
        global_net_flow = sum(weighted_flows.values())
        return weighted_flows, global_net_flow

def save_results_to_excel(avg_results, diff_matrices, preference_matrices,
unicriterion_flows, weighted_flows, global_net_flow, yearly_results,
global_net_flow_yearly, diff_matrices_yearly, preference_matrices_yearly):

        file_name = "Quality of Life EU.xlsx"
        with pd.ExcelWriter(file_name, engine="openpyxl") as writer:
                avg_results["Global_NetFlow"] = global_net_flow
                avg_results["Rank"] =
avg_results["Global_NetFlow"].rank(ascending=False, method='dense')
                avg_results.to_excel(writer, sheet_name="Global NetFlows", index=False)

                yearly_results = combined_data.copy()
                yearly_results["Global_NetFlow"] = global_net_flow_yearly
                yearly_results["Rank"] =
yearly_results["Global_NetFlow"].rank(ascending=False, method='dense')
                yearly_results.to_excel(writer, sheet_name="Global_NetFlow_Yearly", index
= False)

for criterion in criteria.keys():
        sheet_name = criterion[:31]

        diff_df = pd.DataFrame(diff_matrices[criterion],
```

```
index=aggregated_data['Country'], columns=aggregated_data['Country'])
        pref_df = pd.DataFrame(preference_matrices[criterion],
index=aggregated_data['Country'], columns=aggregated_data['Country'])
        phi_plus = unicriterion_flows[criterion]["Positive_Flow"]
        phi_minus = unicriterion_flows[criterion]["Negative_Flow"]
        net_flow = unicriterion_flows[criterion]["Net_Flow"]
        weighted_net_flow = weighted_flows[f"Weighted_{criterion}"]

        flow_df = pd.DataFrame({
                'Country': aggregated_data['Country'],
                'Positive_Flow (Φ⁺)': phi_plus,
                'Negative_Flow (Φ⁻)': phi_minus,
                'Net_Flow (Φ)': net_flow,
                'Weighted_NetFlow': weighted_net_flow
        }).sort_values(by='Net_Flow (Φ)', ascending=False)

        diff_df.to_excel(writer, sheet_name=sheet_name, startrow=0, index=True)
        pref_df.to_excel(writer, sheet_name=sheet_name, startrow=len(diff_df) + 5,
index=True)
        flow_df.to_excel(writer, sheet_name=sheet_name, startrow=len(diff_df) +
len(pref_df) + 10, index=False)



        selected_criterion = "C.6 Unemployment"

        diff_df_yearly =
        pd.DataFrame(diff_matrices_yearly[selected_criterion],index=yearly_results["Count
        ry"] + yearly_results["Year"].astype(str), columns=yearly_results["Country"] +
        yearly_results["Year"].astype(str))

        pref_df_yearly =
        pd.DataFrame(preference_matrices_yearly[selected_criterion],index=yearly_results
        ["Country"] + yearly_results["Year"].astype(str),
```

```
        columns=yearly_results["Country"] + yearly_results["Year"].astype(str))

        diff_df_yearly.to_excel(writer, sheet_name=f"{selected_criterion}_Diff_Yearly")
        pref_df_yearly.to_excel(writer, sheet_name=f"{selected_criterion}_Pref_Yearly")



    print(" PROMETHEE results saved with Differences Matrix, Preference Matrix, and Flows!")



def graph_digraph(global_net_flow, country_names, title):
        s = Digraph(title, node_attr={'shape': 'plaintext'})
        s.body.extend(['rankdir = LR'])

        sorted_indices = np.argsort(global_net_flow)[::-1]
        sorted_flows = global_net_flow[sorted_indices
        for i, idx in enumerate(sorted_indices):
                country = country_names[idx]
                score = f"{sorted_flows[i]:.4f}"

                s.node(
                f'node{idx}', f'''<
                <TABLE BORDER="0" CELLBORDER="1" CELLSPACING="0" CELLPADDING="4">
                <TR><TD COLSPAN="2" bgcolor="lightgray">{country}</TD></TR>
                <TR><TD>Net Flow</TD><TD>{score}</TD></TR>
                </TABLE>>'''
                )
        s.edges([(f'node{sorted_indices[q]}', f'node{sorted_indices[q + 1]}') for q in range(len(sorted_indices) - 1)])

pdf_filename = f'graph_{title}.pdf'
```

```
s.render(pdf_filename, format='pdf', cleanup=True)
print(f" Ranking graph '{title}' saved as {pdf_filename}!")

def scatterplot(global_net_flow, country_names, title):
        plt.figure(figsize=(18, 8))
        plt.scatter(range(len(global_net_flow)), global_net_flow, color='red', label="Net
Flows")

        for i, txt in enumerate(country_names):
                plt.annotate(txt,
                                (i, global_net_flow[i]),
                                textcoords="offset points",
                                xytext=(0, 10),
                                ha='center',
                                fontsize=8,
                                rotation=75,
                                bbox=dict(boxstyle="round,pad=0.2", fc="white", ec="gray",
lw=0.5, alpha=0.7))

plt.xticks([], [])
plt.title(title, fontsize=14, fontweight='bold')
plt.ylabel("Net Flows (Φ)", fontsize=12)
plt.legend()
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()




def main():

        xavg = aggregated_data.iloc[:, 1:].to_numpy()
        pavg = compute_thresholds(aggregated_data)
```

```
        diff_matrices = compute_differences(xavg, c)
        preference_matrices = compute_preference_matrix(diff_matrices, pavg[0],
pavg[1], pavg[2], f)
        unicriterion_flows = compute_flows(preference_matrices)
        weighted_flows, global_net_flow = compute_weighted_flows(unicriterion_flows, w)
        print(pavg)

        x_yearly = combined_data.iloc[:, 2:].to_numpy()
        p_yearly = compute_thresholds(combined_data.iloc[:, 2:])
        print("Debug: q_yearly values (indifference threshold):\n", p_yearly[0])
        print("Debug: p_yearly values (preference threshold):\n", p_yearly[1])
        print(" s value", p_yearly[2])
        diff_matrices_yearly = compute_differences(x_yearly, c)
        preference_matrices_yearly = compute_preference_matrix(
                diff_matrices_yearly,
                p_yearly[0],
                p_yearly[1],
                p_yearly[2],
                f
)



    selected_criterion = "C.6 Unemployment"
    print(f" 🔹 Difference Matrix for {selected_criterion} (Yearly):\n",
diff_matrices_yearly[selected_criterion])
    print(f" 🔹 Preference Matrix for {selected_criterion} (Yearly):\n",
preference_matrices_yearly[selected_criterion])


unicriterion_flows_yearly = compute_flows(preference_matrices_yearly)
weighted_flows_yearly, global_net_flow_yearly =
compute_weighted_flows(unicriterion_flows_yearly, w)
```

```
country_names = aggregated_data['Country'].tolist()
country_names_yearly = combined_data[['Country', 'Year']].apply(lambda row:
f"{row['Country']}{row['Year']}", axis=1).tolist()

graph_digraph(global_net_flow, country_names, "Global NetFlow Ranking")
scatterplot(global_net_flow, country_names, "PROMETHEE II Net Flow")

graph_digraph(global_net_flow_yearly, country_names_yearly, "Global Netflows Ranking
Yearly")
scatterplot(global_net_flow_yearly, country_names_yearly, "PROMETHEE II Net Flows
Yearly")

save_results_to_excel(aggregated_data, diff_matrices, preference_matrices,
unicriterion_flows, weighted_flows, global_net_flow , combined_data ,
global_net_flow_yearly,diff_matrices_yearly, preference_matrices_yearly )

if __name__ == '__main__':
        main()
```

Appendix D: Normalisation Weights Python Script

```
import pandas as pd

file_path = "Quality Of Life (Απαντήσεις).xlsx"
df = pd.read_excel(file_path)
df_cleaned = df.iloc[:, 1:-1]
```

```
average_scores = df_cleaned.mean()
normalized_weights = average_scores / average_scores.sum()
weights_df = pd.DataFrame({
                "Criterion": normalized_weights.index,
                "Weight": normalized_weights.values
})

print(weights_df)
```

Appendix E: Excel file (Data for every country, indicator)

Average Scenario:

Country	C.1 Median_Income	C.2 GDP_Per_Capita	C.3 Ends_Meet	C.4 H_Deprivation	C.5 H_Overcrowding	C.6 Unemployment	C.7 Unexpected_Expenses	C.8 Arrears	C.9 Criminality	C.10 Pollution
DK	30502.6	32843	3.49	15.625	8.95	5.54	23.82	7.05	7.6875	7.6625
EL	8346.1	13379	38.27	13.55	28.35	18.77	49.46	43.9	15.4875	20.4625
ES	15137	16790	11.35	16.625	6.01	16.84	36.92	11.62	11.3875	10.1625
HU	5759.5	7769	13.19	22.7	27.38	4.68	43.26	14.88	7.8625	12.9375
LU	38284.4	41270	3.71	16.575	7.47	5.75	21.22	5.8875	12.1	15.175
NL	25013.3	26458	2.59	15.25	3.87	4.75	19.79	4.07	16.75	14.2625
PT	9861.3	13788	14.2	27.8	10.12	8.68	34.82	7.51	8.0625	13.6375
RO	3797.2	6885	14.66	10.975	45.32	5.43	49.02	16.94	11.1	14.1
SE	25588.4	30795	3.32	7.0625	14.89	7.47	20.25	5.54	12.2375	6.5875
SK	7866.3	10358	9.83	5.9625	34.39	8.02	31.9	8.09	5.9	9.7625

Figure 57: Data, Average Scenario

Yearly Scenario:

Country	Year	C.1 Median_Income	C.2 GDP_Per_Capita	C.3 Ends_Meet	C.4 H_Deprivation	C.5 H_Overcrowding	C.6 Unemployment	C.7 Unexpected_Expenses	C.8 Arrears	C.9 Criminality	C.10 Pollution
DK	2014	27861	30940	4.4	15	8.2	6.9	28.5	7.9	9.1	6.8
DK	2015	28364	31410	3.5	16.1	8.1	6.3	26.5	6.3	7.7	6.6
DK	2016	28665	31930	3.2	15.9	8.2	6	24.5	5.8	8.4	6.8
DK	2017	29383	32400	3.4	14.9	8.6	5.8	25.1	6	7.8	7.9
DK	2018	30104	33150	3.5	16.4	9.2	5.1	25.2	8.7	7.4	8.1
DK	2019	30717	33500	3.2	14.9	10	5	22.9	7.3	7.5	8.4
DK	2020	30681	32810	3.2	16.8	9.5	5.6	22.7	7.7	7.3	9.4
DK	2021	32088	34820	2.8	15.625	9.2	5.1	19.5	6.3	7.6875	7.6625
DK	2022	33260	33710	3.2	15.625	9.8	4.5	20.2	6.7	7.6875	7.6625
DK	2023	33903	33760	4.5	15	8.7	5.1	23.1	7.8	6.3	7.3
EL	2014	7680	12340	39.5	13.7	27.4	26.5	51.8	46.4	16.1	23.2
EL	2015	7520	12510	38.2	15.1	28.1	24.9	53.4	49.3	12.8	19.2
EL	2016	7500	12640	40.6	14.7	28.7	23.6	53.6	47.9	11.8	19.6
EL	2017	7600	12920	39.9	13.5	29	21.5	52.7	44.9	13.8	20.3
EL	2018	7863	13210	38.2	12.9	29.2	19.3	50.4	43	13.5	20.5
EL	2019	8195	13610	37.7	12.5	28.7	17.3	47.8	41.4	16.9	20.2
EL	2020	8781	12970	36.1	12.5	29	16.3	50.7	36.9	18.1	20.2
EL	2021	8752	13660	39	13.55	28.5	14.7	46.3	36.4	15.4875	20.4625
EL	2022	9520	14790	36.8	13.55	28	12.5	43.6	45.5	15.4875	20.4625
EL	2023	10050	15140	36.7	13.5	26.9	11.1	44.3	47.3	20.9	20.5
ES	2014	13269	15760	17.5	17.1	5.3	24.5	42.7	12.5	11.9	10.2
ES	2015	13352	16250	14.8	15.2	5.5	22.1	39.8	11.7	10	10.3
ES	2016	13681	16660	16.6	15.9	5.4	19.6	38.7	10.6	10.3	10.1
ES	2017	14203	17110	9.5	11.5	5.1	17.2	36.6	9.3	8.7	8.2
ES	2018	14785	17360	10.4	15.9	4.7	15.3	35.9	9.4	10.9	10.1
ES	2019	15015	17460	7.8	14.7	5.9	14.1	33.9	8.1	11.6	9.9
ES	2020	16043	15750	10	19.7	7.6	15.5	35.4	13.5	14.1	11.8
ES	2021	15892	16800	8.8	16.625	6.4	14.9	33.5	14.3	11.3875	10.1625
ES	2022	16814	17280	8.7	16.625	6.6	13	35.5	13.2	11.3875	10.1625
ES	2023	18316	17470	9.4	23	7.6	12.2	37.2	13.6	13.6	10.7

Figure 58: Data, Yearly Scenario part a

Country	Year	C.1 Median_Income	C.2 GDP_Per_Capita	C.3 Ends_Meet	C.4 H_Deprivation	C.5 H_Overcrowding	C.6 Unemployment	C.7 Unexpected_Expenses	C.8 Arrears	C.9 Criminality	C.10 Pollution
HU	2014	4512	6460	22.8	26.9	41.9	7.7	75.9	24.5	13.9	15
HU	2015	4556	6720	19.1	25.4	41.1	6.8	72.2	21.7	10.6	14.8
HU	2016	4768	7030	17.1	26.7	40.4	5.1	50.8	19	9.7	12.8
HU	2017	4988	7400	15.7	24.8	40.5	4.2	31.5	15.7	7.2	12.5
HU	2018	5424	7720	12.1	22.5	20.1	3.7	33.3	12.8	4.8	9.7
HU	2019	5852	8090	8.6	22.3	20.3	3.4	33	11.2	5.3	12.4
HU	2020	6478	7990	10.7	20.4	19	4.2	35.7	11.6	5.3	11.7
HU	2021	6619	8380	9.8	22.7	17.9	4	34.8	11.2	7.8625	12.9375
HU	2022	6975	8960	8.1	22.7	17	3.6	33.9	10.3	7.8625	12.9375
HU	2023	7423	8940	7.9	12.6	15.6	4.1	31.5	10.8	6.1	14.6
LU	2014	34320	39960	3.9	15	6.7	5.9	23.8	4.9	16.8	15.4
LU	2015	35270	40130	4.2	14.4	6.8	6.7	23	5.2	14.9	17.1
LU	2016	32841	40190	4.2	18.7	8.1	6.3	21.9	6.6	12.2	16.1
LU	2017	36315	40540	5.1	17.4	8.3	5.5	20.4	3	12	18.5
LU	2018	34472	41270	4.7	18.3	8.4	5.6	19.7	5.8875	11.3	15
LU	2019	36354	41470	3.7	15.4	7.1	5.6	16.7	5.8875	11.2	15.2
LU	2020	37844	39200	3.8	15.4	8.5	6.8	22.5	4.9	11	14.6
LU	2021	42482	42340	3.8	16.575	7.7	5.3	21.3	6.6	12.1	15.175
LU	2022	45310	43920	1.5	16.575	5.7	4.6	18.8	7.2	12.1	15.175
LU	2023	47636	43680	2.2	18	7.4	5.2	24.1	8.7	7.4	9.5
NL	2014	20891	25390	4.3	15.8	3.5	7.4	23.7	6.3	18.5	15
NL	2015	21292	25700	3.3	15.7	3.3	6.9	22.9	5.7	17.4	13.2
NL	2016	22733	25830	3	16.3	4	6	22.5	5	16.9	13.2
NL	2017	23561	26250	3.2	13.5	4.1	4.9	20.7	4.6	15	12.9
NL	2018	24016	26720	2.2	15.8	4.1	3.8	21.5	3.8	17.5	16
NL	2019	24612	26940	2.9	14.7	4.8	3.4	21.9	4	16.3	14.9
NL	2020	25801	25590	1.8	14.8	4.8	3.8	19.1	3.2	15.7	14.2
NL	2021	28441	26640	1.6	15.25	3.4	4.2	15.1	2.6	16.75	14.2625
NL	2022	29537	27740	1.9	15.25	2.9	3.5	14.6	2.9	16.75	14.2625
NL	2023	29249	27780	1.7	15.4	3.8	3.6	15.9	2.6	16.7	14.7

Figure 59: Data, Yearly Scenario part b

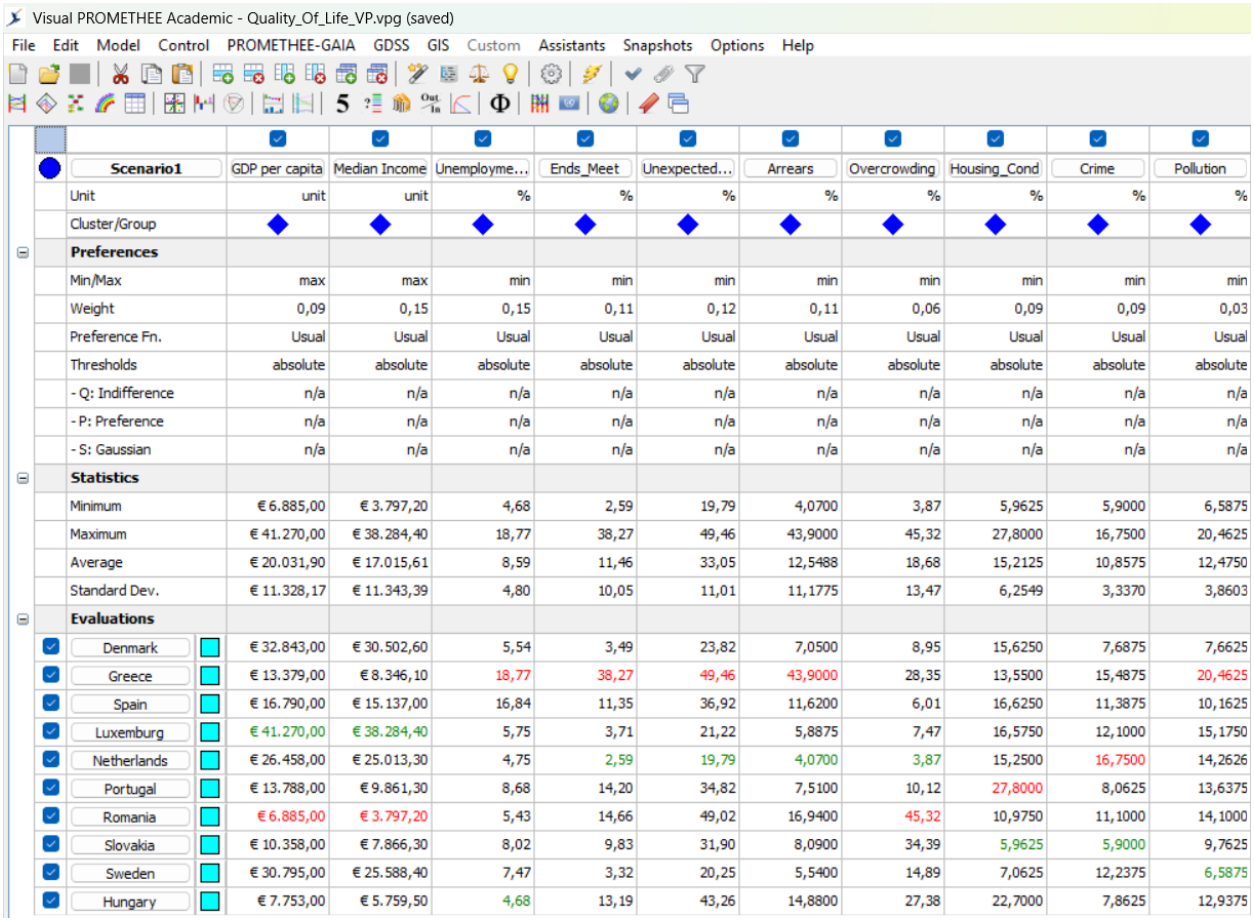
Country	Year	C.1 Median_Income	C.2 GDP_Per_Capita	C.3 Ends_Meet	C.4 H_Deprivation	C.5 H_Overcrowding	C.6 Unemployment	C.7 Unexpected_Expenses	C.8 Arrears	C.9 Criminality	C.10 Pollution
PT	2014	8229	12670	23.3	32.8	10.3	14.1	42.2	12	11.6	13.7
PT	2015	8435	12960	20.6	28.1	10.3	12.6	40.7	10.2	10.5	14.2
PT	2016	8782	13270	17	30.5	10.3	11.2	38.3	9.3	7.8	13.1
PT	2017	9071	13510	15.2	25.5	9.3	9	36.9	7.7	7.9	13.2
PT	2018	9346	13860	13.3	26.9	9.6	7.1	34.7	6.6	6.5	12.9
PT	2019	10023	14340	11.1	24.4	9.5	6.5	33	5.8	6.7	13.5
PT	2020	10800	13440	9	25.2	9	6.9	30.8	5.4	6.6	13.2
PT	2021	11089	14130	11.3	27.8	10.6	6.7	31.2	6.8	8.0625	13.6375
PT	2022	11014	14810	11.2	27.8	9.4	6.2	29.9	6.1	8.0625	13.6375
PT	2023	11824	14890	10	29	12.9	6.5	30.5	5.2	6.9	15.3
RO	2014	2155	5180	21.7	13.6	49.4	6.8	52.7	23.9	14.5	16.1
RO	2015	2315	5490	20.2	12.8	49.7	6.8	51.4	19.6	13.1	15.3
RO	2016	2448	5770	18.1	13.3	48.4	5.9	54.5	19.7	14.1	14.5
RO	2017	2742	6480	14.7	11.1	47	4.9	52.5	17.3	11.3	14.6
RO	2018	3284	7110	13.4	10.1	46.3	4.2	45.9	16.5	11.5	14.8
RO	2019	3851	7420	12.4	9.4	45.8	3.9	44.3	15.4	9.6	13.5
RO	2020	4267	7200	14.8	10	45.1	5	47.3	14.8	8.8	12.3
RO	2021	4830	7720	11.1	10.975	41	5.6	47.3	9.4	11.1	14.1
RO	2022	5512	8140	10.4	10.975	40.5	5.6	47.9	18.4	11.1	14.1
RO	2023	6568	8340	9.8	7.5	40	5.6	46.4	14.4	5.9	11.7
SE	2014	25838	29670	3.8	7.7	12.7	8	20.4	6.4	10.7	8.1
SE	2015	25306	30480	3.2	7.7	13.9	7.4	19.8	6	10.9	7.6
SE	2016	25164	31010	3.1	7.4	14.4	7	20.7	5.4	12.7	6.3
SE	2017	25376	31090	2.9	7	13.5	6.7	19.7	5.1	13	6.8
SE	2018	25540	31130	3.5	7.8	15.2	6.4	20.2	4.7	14.4	6.4
SE	2019	24474	30960	3.3	7	15.6	6.8	20.5	4.8	13	6.6
SE	2020	24700	29890	3.6	7.1	15.6	8.3	19.8	4.9	13.8	5.9
SE	2021	25498	31260	3.2	7.0625	15.7	8.9	18.4	5.2	12.2375	6.5875
SE	2022	26738	31540	3.1	7.0625	15.9	7.5	21.2	6.2	12.2375	6.5875
SE	2023	27250	30920	3.5	4.8	16.4	7.7	21.8	6.7	9.4	5

Figure 60: Data, Yearly Scenario part c

Country	Year	C.1 Median_Income	C.2 GDP_Per_Capita	C.3 Ends_Meet	C.4 H_Deprivation	C.5 H_Overcrowding	C.6 Unemployment	C.7 Unexpected_Expenses	C.8 Arrears	C.9 Criminality	C.10 Pollution
SK	2014	6809	8900	12.6	7	38.6	13.2	38.9	8.3	8.7	12.2
SK	2015	6930	9190	11.7	6.3	37.8	11.5	36.7	7.5	7.3	11.2
SK	2016	6951	9470	10.1	6.2	37.9	9.7	37.9	7.5	6.9	9.3
SK	2017	7183	9950	8.1	6.7	36.4	8.1	34.6	7.4	6.2	10.7
SK	2018	7462	10340	8.4	5.1	35.5	6.5	31.5	9.9	4.8	9.6
SK	2019	8119	10710	9.4	5.7	34.1	5.8	30	10.2	5.6	9.5
SK	2020	8703	10720	8.6	4.9	30.1	6.7	26.1	6.7	4.3	9.8
SK	2021	8473	11170	9.3	5.9625	31.2	6.8	27	6.3	5.9	9.7625
SK	2022	8819	11740	9.7	5.9625	31.8	6.1	27	8.3	5.9	9.7625
SK	2023	9214	11390	10.4	5.8	30.5	5.8	29.3	8.8	3.4	5.8

Figure 61: Data, Yearly Scenario part d

Appendix F: Visual PROMETHEE file



Visual PROMETHEE Academic - Quality_Of_Life_VP.vpg (saved)

File Edit Model Control PROMETHEE-GAIA GDSS GIS Custom Assistants Snapshots Options Help

Scenario1	GDP per capita	Median Income	Unemployme...	Ends_Meet	Unexpected...	Arrears	Overcrowding	Housing_Cond	Crime	Pollution
Unit	unit	unit	%	%	%	%	%	%	%	%
Cluster/Group	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆
Preferences										
Min/Max	max	max	min	min	min	min	min	min	min	min
Weight	0,09	0,15	0,15	0,11	0,12	0,11	0,06	0,09	0,09	0,03
Preference Fn.	Usual	Usual	Usual	Usual	Usual	Usual	Usual	Usual	Usual	Usual
Thresholds	absolute	absolute	absolute	absolute	absolute	absolute	absolute	absolute	absolute	absolute
- Q: Indifference	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
- P: Preference	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
- S: Gaussian	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Statistics										
Minimum	€ 6.885,00	€ 3.797,20	4,68	2,59	19,79	4,0700	3,87	5,9625	5,9000	6,5875
Maximum	€ 41.270,00	€ 38.284,40	18,77	38,27	49,46	43,9000	45,32	27,8000	16,7500	20,4625
Average	€ 20.031,90	€ 17.015,61	8,59	11,46	33,05	12,5488	18,68	15,2125	10,8575	12,4750
Standard Dev.	€ 11.328,17	€ 11.343,39	4,80	10,05	11,01	11,1775	13,47	6,2549	3,3370	3,8603
Evaluations										
Denmark	€ 32.843,00	€ 30.502,60	5,54	3,49	23,82	7,0500	8,95	15,6250	7,6875	7,6625
Greece	€ 13.379,00	€ 8.346,10	18,77	38,27	49,46	43,9000	28,35	13,5500	15,4875	20,4625
Spain	€ 16.790,00	€ 15.137,00	16,84	11,35	36,92	11,6200	6,01	16,6250	11,3875	10,1625
Luxemburg	€ 41.270,00	€ 38.284,40	5,75	3,71	21,22	5,8875	7,47	16,5750	12,1000	15,1750
Netherlands	€ 26.458,00	€ 25.013,30	4,75	2,59	19,79	4,0700	3,87	15,2500	16,7500	14,2626
Portugal	€ 13.788,00	€ 9.861,30	8,68	14,20	34,82	7,5100	10,12	27,8000	8,0625	13,6375
Romania	€ 6.885,00	€ 3.797,20	5,43	14,66	49,02	16,9400	45,32	10,9750	11,1000	14,1000
Slovakia	€ 10.358,00	€ 7.866,30	8,02	9,83	31,90	8,0900	34,39	5,9625	5,9000	9,7625
Sweden	€ 30.795,00	€ 25.588,40	7,47	3,32	20,25	5,5400	14,89	7,0625	12,2375	6,5875
Hungary	€ 7.753,00	€ 5.759,50	4,68	13,19	43,26	14,8800	27,38	22,7000	7,8625	12,9375

Figure 62: Visual PROMETHEE File

PROMETHEE Flow Table

Rank	action	Phi	Phi+	Phi-
1	Netherlands	0,5000	0,7500	0,2500
2	Denmark	0,4778	0,7389	0,2611
3	Sweden	0,4244	0,7122	0,2878
4	Luxemburg	0,3711	0,6856	0,3144
5	Slovakia	-0,0200	0,4900	0,5100
6	Spain	-0,1822	0,4089	0,5911
7	Portugal	-0,2289	0,3856	0,6144
8	Hungary	-0,2378	0,3811	0,6189
9	Romania	-0,4311	0,2844	0,7156
10	Greece	-0,6733	0,1633	0,8367

Figure 63: Visual PROMETHEE Net Flows, Average Scenario, Usual Function